# The Free Colony

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[Dedication goes here.]

**Chapter One — Clause 9.4.2**

Forty-seven people in a room designed for thirty, and not one of them was happy about it.

Cole Maren sat in a chair that cost someone at Asterlyne a lot of points in a procurement spreadsheet—cheap enough that nobody would notice the line item, expensive enough to signal that this process mattered. Professional. Controlled. The kind of seating that said: *We take you seriously, but not too seriously, because if you were serious people, you wouldn't be here.*

The waiting room smelled like new plastic and the desperation everyone was pretending not to emit. Fluorescent ceiling panels buzzed at a frequency Cole could feel in his molars. A tablet on the low table in front of him displayed the charter—all 247 pages of it, not the 200 they'd claimed in the recruitment materials. He'd started counting on page one and hadn't stopped.

Around him, people were talking. Not talking, really. Performing the social shapes of talking—the woman to his left asking the woman next to her whether she'd done the medical screening, both of them knowing the answer was yes, both of them needing the distraction of words. The man in the corner—contractor's shoulders, calloused hands, not the type to volunteer for anything unless there was something urgent he was running from—staring at a tablet with the concentration of someone who'd decided not to read.

Cole read.

He had the luxury of being methodical. Nobody here knew him. Nobody was going to ask why he wasn't pacing the way the young guy near the door was pacing, or networking the way the two botanists in the corner seemed to be networking, trading names and institutional affiliations like they were at a conference instead of a one-way ticket off Earth. Cole could sit. Could read. Could let the fluorescent light strip the personality out of his face until he looked like he belonged here—which is to say, he looked like someone with nothing better to do.

The contract was written in English, which was kind. It could have been written in legalese—that subspecies of English that used all the same words but rearranged them to prevent understanding. But this contract was designed to be read. Just not *by* the signatories. The structure was forward-facing, user-friendly, the language was clear, the sections labeled sensibly. A person reading it in good faith would arrive at a different understanding than a person reading it carefully.

Cole read carefully.

Section 1: Organizational Charter of the Free Colonial Initiative, Parcel 16b Settlement.

All colonists agreed to the terms without qualification, without amendment, without negotiation. Fine. Nobody negotiated with Asterlyne. That wasn't the system.

Sections 2–5: the basic infrastructure. Settlement boundaries. Resource allocation. Command structure. A temporary governance council elected by majority vote, authority subordinate to the Company Safety Officer, who retained final decision-making on all matters of environmental management and resource distribution.

The woman next to Cole glanced down at his tablet. He felt her looking. He didn't move, didn't look back, just kept reading. She went back to not reading hers.

Section 6: Ownership and Land Rights.

This was where people usually started skimming.

*Upon successful establishment of self-sustaining agricultural capacity and the passage of five (5) Earth years, title to Parcel 16b shall transfer to the colonist collective, contingent upon maintenance of charter*

*compliance and population threshold stabilization.*

Cole read that sentence three times.

Population threshold stabilization.

He flipped to the appendices. Nobody flipped to the appendices. The appendices were where you found the definitions that made you understand what you'd actually agreed to.

Clause 9.4.1: Definition of "colonist collective."

*Ownership stake in Parcel 16b (upon transfer) shall be distributed equally among all original signatories to this charter present and alive on Year Five, Day One, beginning at 0000 hours UTC+7. Signatories who voluntarily depart the settlement, fail to renew charter consent, or are deceased shall forfeit their stake and have such stake distributed pro-rata among remaining signatories.*

Okay. Straightforward. Forty-seven of them now. If someone left early, the pie got smaller, but their slice got bigger. People understood that math. It was a motivator—every person who stayed made everyone else's stake slightly larger. Asterlyne was good at architecture like that. Creating incentives that looked aligned but actually severed the group's interest from their survival.

Clause 9.4.2: Stabilization Threshold.

*Transfer of title shall occur only if, at Year Five Day One, no fewer than fifty-one percent (51%) of original signatories remain alive and present at the settlement. Should this threshold not be met, title shall revert to Asterlyne Corporation in perpetuity, along with all improvements and resources thereupon. No compensation to colonists or next of kin shall be due.*

Cole set the tablet down.

He looked at it on the table like it had grown teeth.

Forty-seven people. Fifty-one percent of forty-seven was 23.97. Functionally, twenty-four. You needed twenty-four people alive in five years, out of the forty-seven who started, to own anything.

Twenty-three could die. That was the margin.

He picked the tablet back up. Read it again. He hadn't miscalculated—his math was exact, and that was the problem. Nearly half the people in this room could die, and the contract would call it a success.

Cole looked up from the screen.

The contractor was still not reading his tablet. The woman next to him was reading the same page she'd been reading five minutes ago. The botanists were talking about soil chemistry, their voices settling into the kind of comfortable technical discussion that meant they'd probably worked together before, probably had a history that didn't start in this room. Near the window, a younger guy—couldn't be more than twenty-two, twenty-three—was rereading the same section over and over, his leg bouncing with the kind of focused anxiety that suggested he was trying very hard to understand something he was afraid to understand.

Cole counted the room. There were the two botanists. The contractor. The woman next to him. The bouncing kid. A doctor, fifties, calm, reading with the measured attention of someone used to processing medical literature. A small cluster of three in the corner who seemed to know each other—energy efficient, probably academic, the type who came to this because they'd burned through their funding and a charter settlement sounded like a field opportunity. Four people he hadn't classified yet. The facilitator, who was polished and bored in exactly the way that suggested she'd watched forty-seven batches of desperate people sign away their rights to land they'd never own.

Forty-seven faces.

Cole looked at each one and asked himself the same question: How many of these people will be dead five years from now?

Not will choose to leave. Not will want to go home. *Will be dead.*

The contractor had an old scar on his jaw that suggested he was good at surviving. That was one point in his favor. The bouncing kid looked like he couldn't survive a bad meal, but youth was its own strategy—faster healing, better reflexes, less likelihood of pre-existing conditions. The doctor had calloused fingertips that came from years of something—surgery, probably. Surgical hands were attached to people who made complicated decisions under pressure. That usually meant longevity.

He was still counting when the facilitator stood.

"We're ready to begin," she said. Her name was Melissa. It was on her badge. She said it the way you'd say: *We're ready to start the experiment.*

They were ushered down a hallway into a smaller room with individual signing stations—corporate ergonomics, each one separated by a divider so nobody had to watch anyone else sign. Privacy theater. Nobody needed privacy to sign away their rights, but it was more comfortable for everyone if they pretended they did.

Cole took his station. His tablet displayed the full charter, pre-populated with his name: COLE MAREN. His background check was complete. His medical clearance was already noted. His initial investment of fifty thousand credits—everything he'd been saving, everything he still owed on the loan that made the fifty thousand possible—was documented. His signature would make it binding.

He didn't sign immediately.

He ran through the math again.

Twenty-four alive. Fifty-one percent. Year five. Five years from day one until the clause triggered. Not five years from now—five years from the day they landed, from the moment the shuttle touched down on Parcel 16b and they became subject to the full weight of the charter.

In five years, Cole would be forty-three. Lily was nine. He'd had custody three years ago, back when the logistics job was still a logistics job and not a series of contract positions that paid less each quarter. He'd held custody like a trophy he'd won and didn't know how to carry, had let it slip through his hands the way he let everything slip eventually—good at systems, terrible at presence.

The career contracted. The apartment shrank. A two-bedroom became a studio became a room he rented by the week. The visits went from weekly to biweekly to monthly to quarterly, each reduction feeling less like a loss and more like a correction.

His ex-wife had filed for full custody eighteen months ago. She hadn't needed to file hard. He'd already surrendered the space where being a father lived.

He thought about Lily at nine, probably taller now than he remembered. The last visit had been awkward. She'd drawn a horse in colored pencil and asked if he wanted to keep it. He'd said yes. Meant it. Left it on the table when he walked out because the recruiter from Asterlyne was waiting and he needed to close the door on something, anything, to stop the feeling that everything he touched fell apart.

The horse was probably in a recycle bin somewhere.

He'd gotten the recruitment email the next day. It had landed in his inbox like an answer to a prayer he hadn't quite had the courage to articulate.

Cole signed.

He used the stylus on the tablet and dragged his name across the signature box. Official. Binding. Witnessed by timestamp, biometric scan, satellite uplink. Legally immutable.

He'd just bet the forty-three-year-old version of himself that he would be alive on Parcel 16b, one of twenty-four survivors.

He'd bet everything he hadn't already lost—on clause 9.4.2.

On the fact that he could count.

On the fact that if you counted carefully enough, you could make the math work. You could read what was written even when nobody wanted you to read it. You could understand what was being asked of you even when nobody would articulate the question.

Twenty-three people. That's what it cost.

He wasn't hoping nobody died. That would be stupid. People always died. Disease, accident, bad luck, the simple arithmetic of time. On Earth, people died too. At least here, the math was clear.

He walked out of the signing room into the hallway where the other forty-six people were filing out of their own stations, each one carrying the same tablet, each one having just signed the same charter, each one presumably having read the same clause.

Or not. Probably not.

Most of them looked like they'd been reading the user-friendly sections, the parts that promised land, ownership, a fresh start. Most of them weren't looking at the man in the corridor the way Cole was looking at them—the way you might look at a group of assets and try to calculate which ones would hold their value.

The contractor from the waiting room passed him. Scarred jaw, calloused hands. His name tag read POLK, G. He glanced at Cole the way you look at someone when you're deciding if they're competition or just another body in the headcount.

"Hell of a thing," Polk said. Not really to Cole. To the corridor, to the fluorescent light, to the general condition of signing away everything you had left.

He walked on before Cole could answer.

Cole watched him go. Polk hadn't read the contract. You could tell from the way he moved—angry but unfocused, the posture of a man who'd signed on instinct rather than analysis. He thought the danger was the frontier. Didn't know the danger was the fine print.

The corridor emptied. The fluorescent hum settled back into Cole's molars.

He'd signed up for the frontier. He'd actually signed up for the margin—the twenty-three lives between ownership and nothing.

And he was betting everything he had left—which wasn't much, which was the point—on the fact that he could count.

**Chapter Two — Transit**

Three months in recycled air tastes like someone else's exhalation mixed with industrial scrubbers. Cole learned to stop noticing it by week two, which meant he'd learned to stop noticing a lot of things. The Asterlyne transport vessel wasn't designed for comfort. It was designed for efficiency and just enough habitability to keep the cargo alive until delivery.

The crew of forty-seven had settled into the lower decks like sediment. Bunks in columns of four. A common mess that doubled as the orientation room. A medical bay where Haruki Sato—the doctor, the calm one, the man who'd introduced himself by asking everyone their medical history instead of their name—had already organized a basic clinic. Exercise equipment that nobody used because moving in 0.7G made the stomach betray you.

Cole spent most of his time reading.

The charter existed in full digital copy on the ship's network, along with supplementary materials: geological surveys, atmospheric data, agricultural projections, structural schematics. Personnel files. Cole started with the geological surveys and worked outward — atmospheric composition, seasonal modeling, soil chemistry baselines. He read the way he'd always read corporate data: not for what was there, but for what should have been. The surveys were clean. Too clean. Methodology notes referenced comparison datasets that weren't included. Soil analysis tables started mid-sequence, as though earlier measurements had been stripped. The gaps were small and the gaps were deliberate.

It was the kind of thing you'd notice if you knew what missing data looked like. Most people didn't.

Most people were doing what they were doing in the common mess on the forty-fifth day of transit: clustering into smaller groups, forming the soft hierarchies that would become the harder hierarchies once gravity meant something again. The two botanists from the initial signing — colleagues from an Earth research institute, both women, both operating under the assumption that the settlement would have built-in scientific infrastructure — had claimed a table and were planning projects. Three academics who didn't quite know each other yet but recognized the type — the kind of people who'd chased this opportunity because their funding had ended, because they had nothing else to lose, because an off-world colony looked like a second act.

Garrett Polk had gathered about eight people around him at the far table and was talking about freedom. Not freedom as an abstraction. Freedom as a wound.

"...systems that look rational," Polk was saying, his voice the kind that filled a room whether you wanted it to or not, "built by people who've never had to live inside them. You do everything right. Follow the plan. Then you find out the plan was never yours. It was for the people who wrote it."

Eight heads nodding. Cole watched from across the mess and heard the specific bitterness underneath the rhetoric — not ideology, but damage. The anger of people who'd played by rules that turned out to be traps, who'd built things that got taken, who'd signed deals that looked like opportunity and turned out to be architecture. Polk wasn't selling a philosophy. He was selling the feeling of finally being right about who had broken you.

Ren Tsai was at a table alone, reviewing something on her personal tablet that made her frown.

Cole had learned her name from the orientation materials. Engineer. Specialty: structural systems and environmental management. She'd been quiet during the embarkation process but had asked three questions during the first orientation presentation that had made the facilitators visibly uncomfortable: questions about the gap between the projected resource allocations and the documented planetary

volatility, questions about failover systems if primary water processing failed, questions about what happened to a settlement if the Company Safety Officer position became unexpectedly vacant.

The facilitators had given the kind of answers that were technically true and completely unhelpful.

Ren had written down their answers anyway, which meant she understood that documentation mattered more than satisfaction.

Cole carried his meal tray to her table. It was a violation of the soft social rules—you didn't join someone sitting alone unless they invited you. But Cole was good at calculating when rules could be bent.

"Can I sit?" he asked.

Ren looked up. She was older than he'd initially thought—maybe mid-thirties instead of late twenties. Her face was the kind that stayed still when it was thinking, which made the moments when it moved mean something. It moved now.

"Sure," she said.

Cole sat. The chair was the kind engineered for low-G comfort, which meant it felt like sitting on a marshmallow someone had wrapped in steel.

"You asked good questions," Cole said.

"They didn't answer any of them," Ren said.

"That's usually how you know they're good questions."

Ren set down her tablet. Up close, Cole could see what she'd been reading: the resource allocation spreadsheet, compared to what looked like her own calculations. She'd been checking the math. Probably finding it didn't add up.

"What's your background?" she asked.

"Logistics. Inventory management, supply chain optimization, that kind of thing."

"So you count things."

"I keep things from running out."

Ren nodded like this explained something. It did. He could see her recalibrating what she was looking at—not just information anymore, but a problem. And problems needed people who could count.

"Why did you sign up?" she asked.

Cole had an answer prepared for this question. It was true and incomplete: "Earth didn't need me anymore. Out there, I might."

It was the kind of answer that closed conversations gently. Ren didn't push. But she looked at him the way someone looks at a calculation they don't quite believe yet, and he knew she was reading the deflection accurately. She understood that he'd answered a different question than the one she'd asked.

She didn't call him on it.

"I'm Ren," she said, and went back to her tablet.

Cole ate his meal — some kind of protein cultivate that had the texture of foam and the flavor of commitment to nutrition — and watched the common mess reorganize itself. The academics had drifted toward Polk's table, drawn by the gravitational pull of a man who sounded like he had answers. The botanists had

separated, one heading to find Haruki in the medical bay, the other — Dana Wen — settling into a corner with her notebook. Actual paper. She filled pages in handwriting so small it looked like code. Who sat where. Who ate what. Which orientation modules people skipped. She avoided eye contact with the practiced ease of someone who'd learned that being overlooked was safety. Cole noticed her because she was designed not to be noticed.

Haruki was visible through the medical bay hatch, sitting across from a colonist in what had become his daily routine — colonists sat down one by one for conversations that started as medical histories and ended with questions nobody wanted answered. How long without your medication? What happens when resupply stops? He asked them in that measured voice, as though he were merely filling in forms.

Joss Carver — twenty-two, the youngest on the manifest — was panicking and preparing in roughly equal measure. He'd cycle through the charter with his leg bouncing, then pivot to the survival appendices, building checklists, cross-referencing agricultural projections with settlement schematics. He'd watched every orientation video Asterlyne had loaded onto the network. Asked Haruki about field wound care. Asked Ren about water system failures. Over-prepared in the way that young people sometimes are when preparation is the only thing they can control.

Polk was laughing about something at his table. Not a happy laugh. The kind that lived in the space between humor and anger, the sound of a man who'd learned to make jokes about things that had broken him.

By the sixty-third day, Cole had learned the rhythms of the ship well enough to move through it without thinking. He knew which paths connected the medical bay to the agricultural storage without passing through the common mess. He knew when Ren went to the exercise equipment and came back all motion and sharp breathing. He knew that Dana did her heavy work in the agricultural lab in the early hours, when nobody else was awake.

He'd organized his own analysis by then. Built a quiet spreadsheet in a private partition of the ship's network. Listed forty-six names — Garrett Polk already filed from signing day, the man who'd said *Hell of a thing* to the corridor and kept walking, all that anger with nowhere to land. Coded them by skill set, by apparent psychological resilience, by the things they'd admitted in casual conversation about why they'd signed.

It was a monster's work, really. Ranking people by survival probability.

But Cole had stopped caring about being anything other than useful.

Polk found him on the observation deck three weeks before planet-fall. The deck was the one section of the ship that looked out at nothing — vast dark, occasional stars, Parcel 16b visible only as a brightening dot among fixed lights.

Polk was at the deck's single table, writing on a napkin. Not words. Numbers. Cole could see the shape of it from across the space — columns of figures, credit amounts, interest calculations. The arithmetic of debt. Polk was comparing what he owed on Earth against what the charter promised, drawing lines between numbers that didn't connect, running the math the way you'd run it if you were looking for an exit that didn't exist.

Cole watched. Didn't hide it.

When Polk realized he was being observed, he crumpled the napkin. The instinct was immediate — not guilt about the numbers, but guilt about being seen running them. His face rearranged itself into the big, public expression Cole had watched him wear in the mess. But the crumpled napkin was still in his fist.

"You read the whole thing, didn't you?" Polk said. "The charter."

"Yes."

"The appendices. The sub-clauses. All of it."

"All of it."

Polk leaned back. Studied Cole the way he studied people — quickly, with the intelligence he tried to keep underneath the volume. Cole could see the question forming. Not the one Polk was about to ask, but the one underneath it. The one that had been on the napkin.

"Find any way out of here?"

Cole considered the question. The honest answer was complicated. There was no way out — no early exit, no loophole that dissolved the contract, no clause that made the debt calculations on Polk's napkin come out different. But there was clause 9.4.2, which wasn't a way out. It was a way through. Five years. Twenty-four alive. Ownership in perpetuity. Polk didn't know about it. Hadn't read far enough to find it.

"No," Cole said.

Polk's face changed. Not defeat — something closer to relief. He lit up the way people light up when someone confirms what they've been afraid to believe: that the game really is rigged, that the math really doesn't work, that it isn't just them.

"No," Polk repeated, almost to himself. "No, there isn't. There never is." He smoothed the crumpled napkin against his thigh, then pocketed it. "I knew it before I signed. Built a whole business on numbers once. Did everything right. Numbers worked. System didn't." He looked out at the dark. "You run numbers long enough, you learn they only work for the people who wrote the equation."

He stood. He was done. He'd gotten what he came for — confirmation from the one person on the ship who'd actually read the contract that the contract was exactly what he already believed it was: a trap you couldn't calculate your way out of.

Cole watched him leave. Understood three things: Polk hadn't read the charter. Polk thought Cole was a fellow prisoner in resignation. And Polk had no idea that resignation and strategy look identical from the outside but are entirely different machines.

Cole didn't correct him. Didn't mention clause 9.4.2. Didn't tell him there was something in the appendices worth finding.

Some information was only useful if the right person had it.

He turned back to his tablet. The weather projections. The agricultural failure vectors. The skills that would matter in months two and three and twelve when Earth was a memory and the supply drops became theoretical.

He wasn't resigned.

He was building the mathematics of survival.

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The final orientation briefing happened five days before planet-fall. All forty-seven colonists gathered in the common mess. A facilitator—not Melissa from the signing station, someone new, someone even more polished and bored—presented the settlement overview.

Parcel 16b. Coordinates. Atmospheric composition. Seasonal cycles. Agricultural viability index. She said the word *index* with the precision of someone reading from a tablet, which is exactly what she was doing.

"The viability index for this parcel," the facilitator said, "has been measured at eighty-four."

Cole looked up.

Eighty-four.

He didn't need to check his tablet. He'd memorized the data appendix weeks ago — every table, every index, every footnote the supplementary materials had buried behind the recruitment brochure. The viability index in the actual survey data was seventy-two.

Seventy-two. Not eighty-four.

A difference of twelve points. The difference between optimism and actual capability. The difference between a projectable settlement timeline and a question mark.

He looked at Ren across the mess. She was looking at him, had been looking at him, had clearly noticed the discrepancy at the same moment and was watching to see if he'd registered it. When their eyes met, she looked away—not dismissive, just confirming. You saw it. I saw it. The question became: what were they going to do about a lie that was buried in supplementary materials that most people wouldn't read?

The facilitator continued her presentation, and Cole returned to his spreadsheet, to his calculations, to the mathematics of scarcity and survival.

The brochure had promised eighty-four.

The planet, he suspected, would deliver seventy-two.

And the colony would have to make the difference work with four thousand kilometers of distance and five years of solitude and the kind of mathematics that only looked simple until you had to live inside it.

Cole started a new column in his spreadsheet.

He labeled it: *Margin for Error*.

He left the cell empty.

He didn't know yet what number should go there, or even if any number was possible.

But he was going to find out.

**Chapter Three — Parcel 16b**

The descent came in three phases. First, the long brake through atmosphere, the shuttle shuddering and groaning the way metal groans when it's asked to defy physics and mostly wins. Second, the sight of it—planet swelling in the viewport from a dot to a thing, from a coordinate to a world. Third, the silence after the engines cut, when forty-seven people held their collective breath and tried to calibrate what they were actually seeing.

Cole watched the planet resolve from his seat, which had a window if you considered a reinforced polymer panel with limited viewing angles a window. Spiral-trunked trees dominated the landscape, their bark twisted like helical stairs climbing to canopy heights that made Earth timber seem practical. The light came through at an angle that suggested either early morning or some orbital geometry that wasn't standard, and the sky wasn't quite blue. Shifted toward green. Lavender-gray soil showed through gaps in vegetation—not barren, just different, the color of something that had been dead once and stayed that way.

Beautiful. That was the thought most of the forty-seven were having, Cole could hear it in the soft sounds they made—not gasps exactly, more like the audible equivalent of holding still. Beauty was a reasonable response to seeing a world for the first time. Cole understood beauty. He could calculate the wavelengths. But he'd trained himself not to *feel* it as a category of data that mattered. Beauty made you make different decisions. Beauty made you take risks.

He'd already calculated enough risks.

"Descent protocol," the pilot announced, her voice crackling through the shuttle's speakers. "Ground contact in approximately four minutes. Secondary teams, confirm your equipment manifest."

Ren Tsai, who'd been sitting three rows behind Cole, was already moving—engineer's efficiency, not tourist curiosity. She was checking seals, verifying that the environmental equipment cases had survived transit intact. Haruki, the doctor, was walking the narrow aisle checking on colonists, making sure nobody was about to vomit or panic or do something that would make the landing harder. The contractor—Polk—was looking at the trees with an expression Cole couldn't quite read. Not afraid. Not happy. Something in between. Maybe it was relief. Maybe it was the look of a man realizing he'd traveled as far as you could travel and the things that broke him had all stayed on Earth.

The shuttle touched down at 0847 hours, local time. The notification came through Cole's tablet: *Parcel 16b Settlement Zone. Atmospheric pressure nominal. Radiation levels acceptable. Welcome.*

Welcome. As if the planet cared.

The descent ramp lowered, and the air that came up into the shuttle was wrong in three ways simultaneously. First, there was the ozone smell—sharp, electric, the kind of smell you got from high-altitude storms. Second, something sweet underneath it, something organic, rotting, alive all at once. Third, nothing. No sound. No birds, no insects, no ambient fauna noise. Just the hum of the shuttle's idle systems and forty-seven people breathing in unison.

Cole was the fifth one down the ramp.

The soil under his boots was springy—higher moisture content than the surveys suggested, or the vegetation had worked out a water table management system that kept things damp near the surface. The air was warmer than he'd expected, which was interesting because the thermal data had been pretty clear about day-night oscillation. This was either early in the warming cycle or the measurements had been taken at a different time of year and the planet's orbital dynamics created seasonal variance the viability index hadn't mentioned.

None of that was his immediate problem.

His immediate problem was the equipment manifest, which was being offloaded by a series of industrial loaders that had their own power sources and protocols and had all been checked by people in Asterlyne facilities who'd never been to Parcel 16b and would never face the consequences of their calculations being wrong. The manifest showed: four pre-fab shelters, rated for thirty occupants at minimum spacing standards. Forty-seven minus thirty equaled a problem that had been solved by putting bunk beds closer together and hoping the structural engineers had built in safety margins they wouldn't tell you about.

Cole inventoried as the loaders set boxes down:

Medical supply cache—comprehensive list on the side panel, he skimmed it while pretending to do something else. Standard Earth medicines. Standard Earth diagnostic equipment. Quantities calculated for a baseline population of approximately twenty, contingency going to thirty. Forty-seven was not a number anyone at Asterlyne had planned for, or if they had, they'd decided the overage could be solved with mathematics that didn't include redundancy.

Agricultural supplies—seed stock in cryo-stasis, hydroponic nutrients, soil amendment compounds. Everything designed to jumpstart food production in alien soil that behaved according to specifications derived from a single survey. Dana Wen, the botanist, was checking the seed cases against her own manifests, her face doing that thing where it stayed still while something underneath it worked very fast.

Shelter composite and assembly supplies—enough to build the four structures that were currently being arranged in a grid by Polk and a cluster of other colonists who'd decided that starting work immediately was somehow less terrifying than standing still. The composite was a polymer-ceramic blend, rated for temperature oscillation within certain parameters. The parameters had a range. The night temperature data suggested they might breach that range.

Water extraction system—mechanical filters, UV treatment, mineral analysis equipment. The system was designed to process maybe 200 liters per day reliably. Cole had done the mathematics: forty-seven people, basic sanitation protocol, was approximately 100 liters daily. Comfortable margins. But that was assuming a consumption model where people showered on schedule instead of hoarding water, assuming the planet's hydrology was stable, assuming the filters worked the way they were supposed to work on water that had never been sampled in the field.

Assumptions were dangerous.

Ren materialized next to him while he was photographing the equipment layout with his tablet, building his own digital manifest against the official one. She was doing the same thing, Cole could see the photograph-and-verify rhythm.

"Medical allocation is short," she said quietly. Not a question, not a panic. Just a statement of mathematical fact. "About thirty percent of full redundancy."

"By design," Cole said. "Asterlyne was betting on low population stability. Figure if enough of us don't survive the first year, the remaining group becomes manageable within their supply calculations."

Ren was quiet for a moment. She was looking at the horizon, at the forest edge about a kilometer out, at the trees that hummed—not audibly yet, but Cole could feel something in the frequency range that made the back of his neck tense. A vibration. A suggestion of sound that wasn't quite sound.

"They knew," Ren said. "When they sent us out here with insufficient resources, they knew."

Cole had already factored this into his model. "The charter is structured to make us expendable. Until we hit the threshold. After that, we become valuable."

"So we survive not because they want us to, but because they suddenly lose something if we don't."

"Yes."

Ren looked at him. "That's a horrible system."

"It is," Cole agreed. "But it's a system. Systems are predictable. I'll take predictable horror over random chance."

The settlement assembly took most of the daylight hours. By 1600 hours, all four shelters were standing—not livable yet, still needing interior partitioning and equipment hookup, but standing. The colonists had fallen into work patterns naturally: the academics organizing a rough commons area, Polk and three others securing the perimeter with what the viability report called a "deterrent fence" and what Cole recognized as a gesture at safety that wouldn't stop anything actually determined to come through. Haruki was establishing the medical shelter setup. Dana was already into the soil, taking samples, running them through the portable analysis equipment.

Cole was finishing a grid map on his tablet, logging every resource location, every distance calculation, every vector for failure he could preempt with foresight.

The sun—wasn't quite a sun, not in the Earth sense, the light quality was wrong somehow, too green, too sharp at the edges—dropped toward the horizon sometime around 1830 hours. The temperature dropped with it, not gradually but in what felt like discrete steps. Cole checked his tablet. Readings dropping from 24 degrees Celsius at 1810 to 18 by 1830 to 11 by 1850. The rate of temperature loss was significantly steeper than the survey had projected. The survey had shown a gradual evening decline over approximately three hours. This was compression. Acute cooling.

Polk noticed it at the same time Cole did. He looked up from securing a section of perimeter fencing and registered the change, and Cole watched his face do the math in real time—contract promised certain conditions, reality was delivering different ones, those weren't bugs, they were features, and features couldn't be returned.

"Inside," Ren called out, already moving the stragglers toward the assembled shelters. "Everyone move inside the structures. Now."

The tone was different from the Earth tone, the one that had asked good questions during the briefing. This was the tone of someone who'd spent three months reading data and had just watched the data prove incomplete. This was the tone of someone whose job had just become impossible.

Forty-seven colonists filed into four shelters that had been rated for thirty.

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The interior of Cole's assigned shelter—one of the four, himself in a bunk room with seven others—was a study in efficient space-wasting. The bunks were stacked in what was technically three-high but felt more like a very elaborate coffin installation. Minimal floor space. Communal sanitation in a corner that was going to create problems once more than two people decided to use it simultaneously. The shelter's environmental system—basic climate control tied to the main power grid—hummed at a frequency that suggested it was working hard at a task it hadn't been designed for.

Night temperature dropped to 3 degrees Celsius by 2200 hours.

Cole lay on his bunk, in a sleeping configuration that was designed for people whose bodies were smaller than human bodies, and listened to the shelter groan. The structure was expanding and contracting as the temperature shifted. The composite was doing exactly what composite does when you ask it to hold rigidity through thermal stress: it was complaining. Not cracking. Not yet. But the sound was the sound of something being asked to do a job within specification while knowing it didn't actually have the margin.

He could hear the other six bunkmates breathing. Someone three bunks over was crying quietly—not a full breakdown, just the kind of soft sounds someone makes when they're trying to process that they've made a mistake and can't unmake it. Someone else was reading something on a tablet, the light bleeding through the darkness—probably the charter, probably looking for a clause they'd missed, a loophole that would suddenly make this sensible.

Cole didn't sleep. He lay in the dark and listened to the shelter and listened to the planet.

Around 0200 hours, the hum changed.

Not louder. Deeper. The vibration that had been in his neck all day was now in his chest, a kind of resonance that felt like it was coming from the ground. Cole pulled up his tablet and checked the seismic sensors he'd configured in the equipment deployment logs. Nothing. No seismic activity. This wasn't the ground moving.

This was something else.

He got out of his bunk—which took an awkward maneuver, bunks were not designed for dignified exits—and moved to the shelter's single observation port. The view was limited to maybe fifty degrees, but it was enough to see the forest edge.

The trees. The spiral-trunked things that had been beautiful in the descent viewport were different at night. They were still visible—there was a faint bioluminescence, something in the bark that glowed a very pale green—and the hum was coming from them. Not an orchestrated hum, not a unified sound. Individual trees, vibrating at slightly different frequencies, creating an interference pattern that wasn't sound so much as the suggestion of sound, the ghost of music that you felt rather than heard.

And something was moving in the undergrowth.

Not approaching. Cole watched carefully. Whatever it was, it wasn't advancing on the settlement. It was there, in the trees, in the shadows that the bioluminescent glow carved out, and it was *looking*.

Cole had spent three months building a spreadsheet of survival probabilities. Had calculated medical failures and resource depletion and interpersonal conflict cascades. Had modeled every variable the transit data offered.

The viability report said nothing about fauna. Not a word. The planet was supposed to be a soil chemistry problem, a weather problem, a supply chain problem. Nobody had mentioned the possibility that something would be out there in the dark, looking back.

He'd calculated everything except what it felt like to be seen by something that had evolved on a world that had never demanded humans exist on it.

The creature—and he was certain it was a creature, singular, patient, interested—watched from the forest. Cole watched back. Neither of them moved. The hours passed like that, both of them aware of each other's presence through the dark, through the humming trees, through a silence that had nothing to do with either of them.

By 0600 hours, it was gone. The forest was still there, still humming, still glowing faintly in the pre-dawn dark. But the sense of being watched released like a held breath.

Cole went back to his bunk. Lay down. The shelter groaned as the temperature began its slow climb back toward morning, and around him the other colonists stirred in the half-dark, shifting against bunks that hadn't been designed for sleep so much as containment.

He could hear them through the thin walls. Through the partitions. Forty-seven people breathing in a settlement that had been on this planet less than a day and was already learning what the planet thought of

it.

Forty-seven.

He needed twenty-four.

The planet had been here for millions of years, cycling through seasons that nothing from Earth was designed for. They'd been here less than a day. Long enough to learn that the air was wrong, the data was incomplete, the shelters were insufficient, and something in the forest was watching.

Cole closed his eyes and did the only thing he knew how to do.

He counted.

**Chapter Four — The Rules**

By day four, the colony had stopped thinking of the planet as an antagonist and started thinking of it as a puzzle. By day seven, the distinction didn't matter anymore. Both puzzles and antagonists could kill you if you didn't pay attention.

Cole spent the first week building systems. It was what he knew — logistics, flow, allocation, redundancy. Warehouse management on an alien world. The same principles applied: figure out what you have, figure out what you'd burn through, calculate how long before the gap between supply and need started killing people.

Water collection came first. The diurnal temperature shifts were steeper than the survey data had suggested — Ren had measured a twenty-degree Celsius swing in twelve hours, double what the viability index predicted. Cole built a schedule around it: collection during the stable morning window, containers sealed by mid-afternoon when atmospheric particles became more active in the cooling hours. Two people per station. Four-hour rotations. Clear protocols for filter type, storage method, contamination prevention.

Food rationing came second. The seed stock had gone into the lavender soil on day two, and by day five, the results were wrong in a way that didn't have an obvious explanation. Some rows took root and sprouted. Others, planted in identical soil ten meters apart, didn't germinate at all. A third set germinated, grew for two days, then wilted overnight like someone had switched them off. Dana Wen was spending twelve hours a day in the agricultural section, running the same tests on the same soil and getting different results every time. She hadn't explained the inconsistency yet, but the intensity of her focus said she was closing in. Until she did, Cole set three rationing tiers: full allocation for physical labor crews, reduced for technical and administrative, baseline for everyone else. Nobody liked the tiers. That was fine. Liking wasn't the point.

Shelter maintenance rotation came third. The pre-fab composite was warping under the UV cycle — Ren showed him the measurements, the thermal stress patterns that exceeded the rated delta every night. Cosmetic damage so far, but structural failure was trending toward month two. She'd drawn up a reinforcement plan that required redirecting labor from agricultural setup, which meant food production would slide, which meant rationing would tighten ahead of schedule. Everything connected. Pull one thread and three others moved.

Perimeter protocols came last, because Cole hadn't figured out what they were protecting against yet. The fauna in the tree line hadn't approached since the first night. The spore activity Haruki had flagged was nighttime-specific. The instability in the soil chemistry was slow, not acute. So Cole set basic rules: nobody beyond the markers after dark, buddy system for scouting parties, mandatory check-ins every four hours.

Systems. Structure. The mathematics of not dying.

The colonists hated it.

Not all of them. Some understood — the ones with technical backgrounds, the ones who'd read enough of the charter to grasp that Parcel 16b wasn't a homestead, it was a five-year survival problem with a contractual threshold. Ren didn't love the hierarchy but she respected the engineering logic. Haruki followed the protocols without comment, which was its own endorsement.

But at least a dozen of them hadn't come here for a shift schedule. They'd come for freedom — for the promise buried in Asterlyne's recruitment materials, the one that whispered: *Leave the systems behind. Start over. Be something other than what the old world made you.* And here was Cole Maren, two weeks on an alien world, already building the same architecture they'd crossed the void to escape.

Garrett Polk led the resistance. Not formally — Polk didn't organize. He just refused, and other people watched him refuse, and refusing became something you could do.

Cole found him at the eastern water station on day five, filling containers without the pre-filter. When Cole pointed it out, Polk didn't argue immediately. He just looked at Cole with something that went past disagreement into a place Cole recognized.

"The filters are slowing us down," Polk said.

"The filters are preventing us from drinking contaminated water."

"We don't *know* that." Polk set a full container down with the controlled force of a man managing his own hands. "We just think it is. Based on the survey. Same survey that said the shelters would handle the cold. Same survey that said the UV wasn't a problem. Everything they promised us was wrong, but we're still following *their* rules?"

"These are my rules," Cole said.

"That's supposed to make me feel better?"

Cole understood what he was hearing. Polk had built something on Earth — a business, a system, something that should have worked. The numbers had been right and the outcome had been wrong, and Polk had lost everything while following rules someone else wrote. Now Cole was standing in front of him with another set of rules, another set of numbers, another architecture that looked rational and promised survival. Polk wasn't seeing a water protocol. He was seeing a trap dressed up as salvation. The same trap, different planet.

When Cole explained the math behind rationing — the caloric intake calculations, the projected agricultural timeline, the consumption models that kept everyone breathing to the five-year mark — Polk listened. Heard every number. And the logic reinforced everything he already believed: that math was the language people in charge used against people like him. Not because the math was wrong. Because the math was right, and it didn't matter.

"I'm not staying here long enough for contaminated water to matter," Polk said. His voice had gone flat — not angry anymore, just settled into the conviction that the game was rigged either way. "Hoping Earth sends rescue by Year Two. Until then, I'm making this livable, not perfect."

Cole documented the deviation and moved on. But he'd seen it now. The math didn't persuade Polk. The math was the problem.

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Cole was still modeling contingencies when Suri Park ate the fruit.

She was one of the ones he'd filed as useful-dangerous. Late twenties, compact, curious. The kind of person who asked questions rapidly and then acted on the answers before fully thinking them through. Useful in exploration. Dangerous in an environment where exploration was still a luxury.

Day thirteen. She'd walked the southern perimeter, mapping the immediate flora, checking what looked edible against the preliminary agricultural database. Found berries clustered on something between a shrub and a low tree. The database showed them safe. Nutrient profile good. No known toxins.

The database had been built from a single survey conducted during a single visit.

Suri ate one.

The reaction came forty minutes later. She was back at the settlement, documenting her finds, when her hands started to shake. Not from cold, not from fear — from something her nervous system was registering as a profound threat. Pupils dilated. Breathing accelerated. Within an hour, she was seizing — full tonic-clonic, her body rejecting a compound that the database said didn't exist, her immune system fighting something Earth medicine had never catalogued.

Haruki got her to the medical shelter. On her side to prevent aspiration. Muscle relaxant from the pharmaceutical stock — shouldn't have worked on an alien compound, but it slowed the seizure enough to keep her from breaking bones. He watched her through the worst of it with the steady hands of a man who had done this before and would do it again.

By day fourteen, Suri had stopped seizing.

Cole sat with Haruki while they waited for her to wake.

"She followed protocol," Cole said.

"She did." Haruki had his hands folded, still. "The database said those berries were safe. It said that based on readings taken during the survey window. Today is not the survey window."

"So the chemistry changed."

"The chemistry is always changing. On Earth we call it seasonal variation — compound expression shifts with temperature, light exposure, soil chemistry. A plant that's safe in June can be toxic in November." Haruki looked at Suri's sleeping form. "The difference is that on Earth, we've had ten thousand years to learn the calendar. Here, we don't even know the length of a season."

Cole sat with that.

The viability data wasn't wrong. It was incomplete. A single survey had captured a single moment, and they'd been navigating by it as if the planet would hold still. But the planet wasn't holding still. The soil chemistry Dana was tracking, the temperature swings Ren was measuring, the compound variations that had nearly killed Suri — all of it was moving, cycling, shifting on a schedule nobody had mapped.

They'd been reading a still photograph and calling it a map.

He went to find Dana. She was in the agricultural section, three soil samples lined up with a fourth added since morning. Same soil. Different readings.

"It's not random," she said without looking up. "There's a pattern. I don't know what's driving it yet, but the shifts aren't noise. They're signal."

"The berries Suri ate."

"Compound concentrations have shifted since the survey. Enough to cross from safe to toxic."

"How long until you can predict the shifts?"

She looked at him. Her expression was the careful blankness of someone running calculations she wasn't ready to share. "I need more data points. Weeks, maybe. If the pattern holds."

"And if it doesn't."

Dana went back to her samples.

Cole called the settlement together that evening. Not a formal assembly — people gathered by the common area between shelters where they'd been eating meals. He told them about Suri. Told them the database was unreliable. Told them the new protocol: nothing native consumed without explicit clearance from both Dana and Haruki. Everything filtered. Everything tested. Everything assumed dangerous until proven otherwise.

Polk spoke first. "So the rules change again."

"The planet changed the rules," Cole said. "I'm keeping up."

"You're good at rules, Maren. Bet you were good at them back home too."

Cole let it land. Polk wasn't entirely wrong. Cole did think in systems. He did impose structure. He did believe the mathematics of organization was the margin between survival and entropy. What Polk missed was the assumption that Cole enjoyed it.

Ren stood. Not confrontational. Just clear.

"The survey was supposed to answer these questions," she said.

"It did," Cole said. "For a specific moment in time."

"Then we need to build our own answers."

That was the cleanest thing anyone had said since landing. Cole let it settle over the group — the idea that they weren't inheritors of someone else's data anymore. They were the survey now. Whatever they learned, they'd learn by living through it.

The gathering broke apart. People went back to shelters, to tasks, to the quiet work of processing that the planet they'd signed up for was not the planet they were on.

Cole stood at the perimeter as the temperature began its drop. The forest hummed. The soil under his boots was springy, responsive — alive in ways he couldn't see and was only beginning to measure.

The planet had rules.

They just hadn't finished learning them yet.

# Chapter 05 — "Three"

The first body was found at dawn.

Cole heard about it before he saw it. Haruki moved through the settlement with a particular efficiency that meant something had already failed beyond his help. They found the colonist — a man named Lev, late thirties, civil engineer from Lagos — about two hundred meters beyond the perimeter markers, in a clearing where the undergrowth thinned and the humming trees opened to actual sky.

Lev had walked out alone at dusk the day before. The buddy system was in place. The four-hour check-ins were in place. Lev had signed out of neither — just left his workstation warm and walked past the markers as if the rules Cole had spent a week building were suggestions he'd decided not to take. Ren found the gap during evening equipment count, ran the name against the check-in log, came up empty. They'd searched until dark, then stopped. Going into the forest after dark had become instinct-level stupid by now. The spore data Haruki had flagged. The temperature swings Ren had measured. Suri Park seizing on day thirteen. Everyone knew the rules about dark.

The rules existed. Lev had decided they didn't apply to him. Or he'd needed to walk badly enough that it didn't matter.

Cole arrived at the clearing while Haruki was still examining the body. No visible trauma — no wound, no animal sign, nothing a layperson would point to and call a cause. Just a man lying in a field who'd stopped breathing. The smell was already present — not rot, something else. Sweet and wrong, like overripe fruit left in closed air. Haruki looked up from Lev's face and Cole did the quick math of exposure. Eight to ten hours minimum.

"Spore response," Haruki said. His hands were gloved. Everything about his posture said this was not a surprise, which meant it was bad in a specific way. "Allergic cascade. Throat closure, pulmonary edema. The lower temperature at night activates the spore germination. I've seen similar compounds in the soil samples Dana was analyzing."

Cole knelt. Not to touch — Haruki would have objected — but to observe. Lev's lips had a blue tint. His hands were clawed, fingers rigid. There was dried crust in his airway. The man had suffocated, fighting for air the planet wouldn't let him take.

He thought about the transit interviews — Haruki's daily intake sessions, the blood panels, the allergy profiles. Lev had been screened. They'd all been screened. But the thing that killed him didn't have a name in any Earth database. The planet had made something new. A vulnerability no human body had ever encountered, triggered by a temperature shift no human schedule had accounted for.

"He didn't follow the check-in protocol," Cole said.

"No."

"Would it have mattered?"

Haruki was quiet for a moment. "If he'd gone out with a partner, the partner would have noticed the respiratory distress. Gotten him back inside the perimeter before full cascade. Maybe." He looked at the body. "But the spores were in the clearing. Once he was exposed, the timeline was short."

Cole stood. The math was already recalculating behind his eyes, but he didn't let it move his face. Forty-six. The new baseline. Not forty-seven minus one — you couldn't count backward. You had to move forward from the number that remained.

The protocols had failed their first real test. Not because they were wrong — because they assumed compliance. Cole had built a system for people who followed systems, and the first man who didn't was dead.

"We need perimeter fence," Cole said. "Real fence. Not markers."

It took four hours. Not the delicate marking he'd set up during the first week — actual barrier. They used pre-fab shelter paneling torn loose, lashed to the trees with the most stable root systems. It looked desperate and temporary and like an admission that they weren't safe in the place they'd landed. Cole worked alongside everyone else. It was easier to maintain authority if people saw you sweat.

Forty-six.

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The second death came three days later.

Storm system moved in faster than the weather models predicted. Not a hurricane — the planet's atmosphere didn't seem to generate those kinds of systems — but sustained wind that came down like a hand pressing everything flat. One of the pre-fab shelters, the one they'd repurposed as equipment storage, had a structural joint that failed under sustained lateral pressure. The beam didn't drop cleanly. It twisted, caught on the shelter frame, held for a moment like it was deciding, then came down across a colonist who'd been inside retrieving supplies.

Her name was Priya. She'd been an accountant. Cole had liked her because she understood margin — the way numbers compressed when you pushed them too far. She understood the cost of precision.

They recovered her body after the wind died. Haruki said she'd died instantly. Cole wasn't sure if that was true or if Haruki was being kind. He didn't ask for clarification.

The shelter was salvageable. The joint was a design flaw in the pre-fab specifications — load-bearing connection rated for wind speeds fifteen percent below what Parcel 16b actually generated. Cole had caught it during the equipment inventory. He'd calculated that losing ten percent of shelter capacity was statistically acceptable against the labor cost of reinforcing every joint in a shelter designed and manufactured to fail at exactly this pressure. He'd made the choice. Priya had paid for it.

Forty-five.

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The third death was the simplest, and that was almost worse.

A colonist named Vogel — mid-twenties, geologist, the kind of person who saw rock formations the way Dana saw soil chemistry — had been scouting a ridge system to the east during clear morning hours. Not explicitly against protocol. He'd signed out properly, taken a partner who waited at the base while Vogel climbed to get a better angle on the underlying geology. Initiative. The kind of thinking that would matter in month six or month twelve, if they got there.

Vogel slipped. Nothing exotic. The stone was more fractured than his geological eye had read from a distance. His foot went through a crack that looked solid. He tried to catch himself on a rock edge that crumbled. He fell maybe fifteen feet into a ravine that opened up sideways.

His partner heard the impact. Called for help. By the time the search party arrived, Vogel was dead. Broken neck. Not the planet's biology. Not spores or shifting soil chemistry or alien compounds the medical kit couldn't treat. Gravity. Bad footing. A ridge that looked stable to a trained geologist and wasn't.

Simple. Stupid. Human.

Haruki examined the body with the same care he'd brought to Lev's, which was a kind of respect that didn't require words. Cole watched from outside the perimeter of Haruki's work. Haruki's job was to witness death. Cole's job was to count it.

Forty-four.

Three dead in one month. One from the planet's hidden biology. One from equipment designed to fail. One from the kind of accident that happened on Earth every day — construction sites, hiking trails, anywhere gravity met bad luck. The planet didn't need to be exotic to kill them. It just needed them to be human.

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Cole called the colony together at dusk on the day they burned Vogel's body. Not a memorial. Memorials took time they didn't have. He called it what it was — a logistics briefing. He stood in front of forty-three other humans and held the numbers in the air between them.

"Forty-four alive," he said. "Forty-seven landed. Three dead in four weeks. At this rate, we hit nineteen dead by month twelve. That leaves twenty-eight."

He let the number sit.

"The charter clause requires twenty-four original signatories alive at the five-year mark. That's the threshold. Below it, Asterlyne keeps the land. Above it, we own it. Right now, we're on a trajectory that dips below twenty-four before year three."

Someone shifted weight. Someone else breathed out like they'd been holding it.

"Every death from here is a number we can't recover. Every protocol violation, every unnecessary risk, every shortcut — it comes off a margin that's already thinner than it should be."

Ren Tsai stood up. Her face was sharp, engineer's problem-solving mode engaged.

"People aren't margin," she said.

Cole met her eyes. This was the conversation. This was always going to be the conversation.

"They are if you want them to be alive," he said.

The two statements hung in the air — both true, both incomplete, both impossible to reconcile. Ren didn't argue further. She sat down. Not because she'd lost, but because she'd said the thing that needed saying and Cole had said the thing that needed answering, and neither of them could close the gap between the two.

Around the settlement, the split happened in real time. Cole could see it in the faces, the posture, the way people looked at him after Ren sat down. Half of them nodded — the ones who understood that arithmetic was the only honest language left, that the numbers were the wall between them and the planet's indifference. They saw a leader. Someone willing to do the work nobody else wanted to do.

The other half looked at Ren. The ones who needed to believe that survival meant more than subtraction. That the colony was a community, not an equation. They looked at Cole and saw a machine — efficient, necessary, and something you wouldn't want to be left alone with in the dark.

Garrett Polk sat in the back and didn't nod at either of them. His face said he was running his own math, and it didn't match Cole's.

Joss Carver sat near the front, hunched forward, hands gripping his knees. He'd been prepared for this — the orientation videos, the field medicine research, the checklists he'd built during transit. He'd run the

scenarios. But his face said the scenarios hadn't included watching real numbers shrink while standing next to the people they represented.

Nobody spoke after Ren. That was the worst part — not disagreement, but the silence of people who understood the problem and couldn't see a solution that didn't require someone else to die.

The gathering broke apart. People moved back to shelters, to tasks, to the quiet private work of processing that they were forty-four and the margin was twenty, spread across fifty-five months.

In his shelter, Cole pulled the tablet where he kept his running calculations. The numbers were already updated, had been since each death. Three entries deleted. Three rows blanked. He stared at the spreadsheet until the names stopped being names and became positions — labor allocation, caloric draw, skill redundancy, loss impact.

Forty-four.

Twenty-four.

The math was simple. The math was always simple. The problem was that the math was people, and people did things like walk past perimeter markers at dusk because they needed air, or work inside shelters with faulty joints because the supplies were needed, or climb ridges because the geology mattered and the stone looked solid.

He put the tablet down. Lay in the dark. The hum of the trees came through the shelter walls — constant, indifferent, the sound of a planet that had been here for millions of years and would be here long after the last of the forty-four had been subtracted to zero.

Half the colony thought he was a leader. Half thought he was a machine.

Both were partially right. That was the part he couldn't calculate his way out of.

# Chapter 06 — "Ghost Colony"

The scouting party found it by accident.

Ren had led a group west to map potential fresh water sources—the mineral content in their current reserves was shifting and Dana wanted a fallback location. They'd marked a rough grid, moved methodically through the forest, counted steps and measured declination. Standard surveying work. Cole wasn't with them. He was managing shelter reinforcement back at the settlement. He learned about the discovery from Joss Carver, who'd been trailing at the rear of the expedition and moved through the settlement looking like his skeleton wanted to escape his body.

"You need to see it," Joss said. No preamble. "Like, immediately. We're not telling anyone yet. Ren said wait for you."

The site was twelve kilometers west, deeper into the forest than Cole had approved for surveying runs. The first marker was a piece of pre-fab shelter paneling, half-absorbed by creeping vegetation. Then another. Then a small clearing where the spiral trees had been deliberately cut or had failed—hard to tell which—and the undergrowth had reclaimed the space with brutal efficiency.

Four shelter units, collapsed in various states of partial excavation by the planet. The aluminum frames were still visible, still Asterlyne's logo, but older. Previous generation. The manufacturing dates on the frame stampings read seven to eight years old. Parcel 16b had been attempted before.

Cole moved through the clearing without touching anything. Ren watched him process. Joss kept looking at the sky like he expected something to drop from it.

The equipment was mostly useless. Corroded harvest tools. Water collection systems rusted to their component minerals. A drone with its solar cells clouded beyond function. All of it was technically salvageable and all of it was worthless. The planet had spent seven years teaching these machines that staying here meant becoming part of the geology.

There was a shelter that hadn't fully collapsed. The door mechanism had seized, but the structure held. Cole worked the door open with a tool Ren handed him. Inside: dust and organized decay. Medical supplies in waterproof containers. Data tablets. A water filtration system that looked like it had been maintained until maintenance stopped mattering. Food storage units empty and lined with what looked like fungal residue.

And, in a sealed case bolted to the interior wall, a data core.

Cole opened it. The seal had held—environmental protocols working correctly even after seven years of submersion in a hostile atmosphere. The core was a standard Asterlyne model, encrypted and dated. Cole didn't try to access it on the spot. He just held it.

"This is a previous expedition," Ren said. Not a question.

"Fourteen colonists," Cole said. He was reading the manifests on the interior wall, printed on material that had survived the weather better than the metal had. "Came in on charter designation XR-Seven. Launch date seven years, four months ago."

"They failed," Joss said. Not a guess. Fact-finding.

"Completely," Cole said.

He pocketed the data core without asking permission. Ren didn't object. They marked the site's coordinates, made a note in the survey log—casual, like they'd found a previous water source they could revisit—and walked back to the settlement without talking.

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That night, Cole sat in his shelter and accessed the core.

The file structure was organized the way a scientist organized data—chronological, labelled, methodical. Expedition logs. Daily resource inventories. Weather observations. Environmental sampling protocols. Crew manifests. The metadata told him what the manifests on the shelter wall hadn't: charter designation XR-Seven was a data extraction contract. No ownership clause. No 9.4.2. No threshold, no five-year mark, no path to keeping anything. Fourteen people sent to study a planet, report their findings, and wait for retrieval. The retrieval timeline was listed as "at corporate discretion upon completion of primary survey objectives."

Corporate discretion. Two words that turned a research mission into an open-ended sentence.

Elena Vasik was the expedition lead, and she wrote like a scientist—precise, minimal, focused on data that mattered. Her first-month entries had the controlled cadence of someone documenting conditions with the expectation that someone competent would eventually read them and act accordingly. Soil chemistry: fluctuating, unpredictable, tied to variables she hadn't yet isolated. Equipment sufficiency: below threshold for sustained habitation. Pre-fab shelters showing thermal stress within the first week. Medical supplies calculated for a smaller population—which, at fourteen, meant calculated for maybe eight. The same corporate arithmetic that assumed attrition would solve the supply problem.

Three deaths in the first month. Spore response, structural failure, a fall during scouting.

Cole stopped reading. Set the tablet down. Picked it up again.

Three deaths. The same three deaths. The same causes, the same order, the same planet running the same sequence on different people seven years apart. Lev, Priya, Vogel—he could see their faces overlaid on Elena's clinical entries like transparencies. She'd watched the same failures happen. Run the same postmortems. Documented the same useless lessons learned.

Eleven remaining in Elena's colony after Month One. Forty-four in Cole's. Different scales. Same trajectory.

He kept reading.

Elena's Month Two entries changed. Not dramatically—she was too disciplined for that—but the sentence length shortened. Questions appeared where observations had been. She'd started sleeping less, which she noted the way she noted soil pH: as a variable she was monitoring without intervention. The colony's social cohesion was fraying. Two of her people had wanted to signal for retrieval from the start, and the fact that retrieval required corporate discretion hadn't stopped them from arguing for it. Trust was becoming a consumable resource, and they were burning through it faster than food.

Two more deaths in Month Two. One from exposure during an overnight temperature drop the weather models hadn't predicted. One from a water contamination event—mineral content shifted between sampling and consumption. Nine remaining. Elena documented each death with the same clinical precision, but she'd started adding a single line after each entry that wasn't data. Just a name, repeated. As if saying it once in the official record wasn't enough.

Cole understood that impulse. He'd been counting names as numbers. Elena had been counting them as names. He wasn't sure which was worse.

The last entry in Elena's Month Two log was different from the others. Longer. The clinical voice cracked open, and underneath it was something that didn't belong in a research report:

"Asterlyne knows the second-year cycle. They chose not to tell us. I found it in the geological survey metadata, buried in an appendix no one was meant to read. The planet operates on multi-year orbital oscillation. The first year is survivable. The second year is hostile—fundamentally different atmospheric chemistry, accelerated biological activity, temperature extremes outside any parameter we were given. They

documented this. They had the data from orbital surveys conducted before our charter was written. They knew what Year Two would do to us, and they sent us with Year One supplies and Year One protocols and Year One expectations. The retrieval clause says 'at corporate discretion.' They are waiting for the planet to make the decision for them."

Cole closed the file.

He sat in the dark for a long time.

Then he pulled his own data package—the one Asterlyne had provided at charter signing. His own geological survey metadata. His own appendix. He scrolled through it, looking for what Elena had found. Climate modeling. Seasonal projections. Multi-year oscillation patterns. Twenty pages of dense data, and then—nothing. The appendix ended mid-section. The next page was formatted wrong, file-size misaligned with the table of contents. Cole checked the document properties. Seven pages removed. The section on second-year orbital dynamics had been redacted before he signed the charter.

Asterlyne hadn't sent them to an undocumented planet. They'd sent them to a documented planet and deleted the documentation. Known what Year Two would bring. Known the supplies were insufficient. Known the shelters would fail. Known the medical kit was designed for a population that would shrink by design.

The company had run the numbers before Cole ever signed his name.

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The data core sat in Cole's shelter for the rest of the week. He didn't open it again.

Not because he'd finished. Elena's logs continued past Month Two—files labelled through months and seasons that stretched across what must have been years of entries before the collapsed shelters twelve kilometers west became the final record. The story didn't end at Month Two. The clearing told him how it ended.

But the middle was still unread. The part where Elena's colony went from nine to the silence of an abandoned clearing. The part that would tell him what the second-year cycle actually looked like from the inside, what it did to people, what it broke that couldn't be reinforced or rationed or calculated away.

He wasn't ready for that yet. He was rationing Elena the way he rationed everything else—taking only what he could use now, saving the rest for when the math demanded it.

He told no one what the logs contained. Not Ren, who'd been there and deserved to know. Not Haruki, who'd asked about the ghost colony twice since they'd returned and accepted Cole's deflection with the patience of a man who recognized strategic silence. Not the council. Not the colony.

He kept it because knowledge was weight, and he was the only one who'd volunteered to carry it. That was the logic. The cleaner logic underneath was simpler: if he told them about the second-year cycle, about the redacted appendix, about a corporation that had sent fourteen people to a planet it knew would kill them and then sent forty-seven more—he'd have to watch forty-three faces process what he'd already processed alone in the dark. He'd have to manage their fear on top of his own.

He began to study the geological appendix—the truncated version—the way Elena must have studied the complete one. Cross-referencing against Dana's soil sampling data. Looking for the pattern Elena had found in the metadata. The orbital oscillation. The cycle that would turn the planet from survivable to hostile.

Dana's data was already showing the edges of it. Mineral fluctuations that didn't track with local weather. Bacterial colony behavior that suggested a seasonal shift happening on a timescale longer than months. The evidence was there if you knew what to look for, and Cole knew now because a dead woman had written it down in a data core sealed against seven years of atmospheric decay.

Elena Vasik had found the pattern and couldn't survive it. Fourteen people, no ownership clause, no reason for Asterlyne to retrieve them, and a planet that operated on a timeline the company had understood and concealed.

Cole had a target. He had clause 9.4.2. He had Dana's soil mapping and Ren's engineering and Haruki's medicine and forty-four people who were still breathing.

Elena had none of that. And the clearing twelve kilometers west was what none of that looked like at the end.

The question was whether having better math and more people and a contractual endgame changed the fundamental equation. Whether knowing where you'd fail made it possible to fail differently. Or whether you just fell in a new direction, the same speed, hitting different ground at the bottom.

Forty-four alive.

The humming trees didn't answer. They just kept humming.

# Chapter 07 — "Factions"

The colony was breaking, and Ren could feel it the way a structural engineer felt load shift in a foundation—not the crack itself, but the settling that came before.

By month three, people had stopped eating together.

It happened in stages. First the meal rotation shortened—people grabbing rations and returning to their shelters instead of sitting at the communal tables Cole had insisted on during week one. Then the small groups formed. Joss and the two youngest colonists maintaining the water filters on their own schedule, separate from the posted rotation. Dana working alone in the soil lab, ignoring the buddy protocol that existed because a man had died from airborne spores six weeks ago. Four colonists who'd bunked together on the transit ship sleeping and waking as a unit, organizing their labor around each other instead of around the settlement.

And Garrett Polk. Always Garrett Polk.

Ren was an engineer. She thought in load-bearing structures and stress distribution and the invisible mathematics of what held things together. She'd spent fifteen years on Earth designing systems that distributed force so no single point bore too much weight. The colony's social architecture was doing the opposite—concentrating stress at fracture lines and pretending the structure was sound.

She could see it. Cole couldn't. Or wouldn't. He saw the logistics: shelter reinforcement schedules, caloric intake projections, equipment maintenance rotations. All of it optimized. All of it correct. All of it useless if the people inside the system didn't trust the system anymore. Cole managed the colony the way he'd managed a warehouse—inputs, outputs, tolerances. He didn't see that the warehouse was full of people who'd come here because they were done being managed.

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On day nineteen of month three, she found Marcus by the water collection array. He was adjusting a seal that didn't need adjusting, and he stopped when he saw her.

"It's fine," he said before she could ask. "Just checking."

"Dana confirmed flow rates this morning."

He straightened up—the small compensatory movement people made when they were caught, drawing themselves taller as if height could substitute for honesty.

"Garrett's worried about redundancy," Marcus said. "Thinks we should have backup systems. Independent ones. Things that don't need Cole's sign-off every time."

There it was. The framework. Cole equals control. Independent equals freedom.

Ren had heard variations of this from six people now. All of them approached it the same way—obliquely, as though they were discussing engineering philosophy instead of splitting the colony in half.

"Garrett's worried we're going to die following Cole's protocols," Ren said. "That the real issue?"

"Isn't it?" Marcus wasn't aggressive. Just tired. "Three dead in month one. Whatever systems Cole's running, they're not working."

"They're working as well as they can given what we have. Which is not great, but better than the last expedition managed."

"I haven't read anything that's going to convince me one person should decide everyone else's life expectancy," Marcus said. "Freedom's the whole point. We came here to get away from people making decisions for us."

"We came here because we had no other choice and Asterlyne had a planet," Ren said. "The freedom part was the pitch."

Marcus didn't respond. He resealed the collection array with the precision of someone who'd been told what to do and wasn't interested in the reasoning.

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She found Polk at the northwest perimeter.

She'd known where to look because Marcus had been spending his shifts out here, and Marcus didn't go anywhere Polk hadn't already mapped. The clearing was sheltered by spiral trees, and in it sat the skeleton of a signal assembly—half-constructed on a wooden frame, jury-rigged from spare parts that should have been allocated to shelter reinforcement, copper wire meant for Dana's soil equipment, an antenna salvaged from the settlement's comm hub.

Polk saw her and didn't bother looking apologetic. That was the thing about Polk—he'd never learned false compliance. He stood there like a man building something unauthorized who intended to finish it.

"Before you start," he said. "This isn't sabotage. It's insurance."

"You're signaling Earth."

"Yes."

"Knowing that a signal voids the charter for everyone."

"If they even respond." Polk crossed his arms. "More likely they ignore it and we figure out another way off this rock."

"That's not how it works, Garrett. There are no passing ships out here. Asterlyne controls the relay. If a signal goes out, they have legal obligation to respond. And they'll respond with a retrieval crew and a bill for extraction that you'll personally owe for the next—"

"I'm aware of the math." He was looking at her directly, which was rare for Polk. Something in Ren's engineering stare made most people feel like she was calculating their structural integrity. Polk didn't flinch from it. "The math says stay here and die slow, hope your bones mean something in five years. Or call out, face the company, at least have a chance of getting off this planet alive."

"You never planned to stay five years."

It wasn't a question. She'd watched him on the transit ship. Watched him during orientation. Polk had signed the charter the way people sign things when the alternative is worse—not with intention but with momentum. He wanted off Earth. He didn't want this.

"I planned to not be dead," Polk said. "I planned to see my daughter again. I didn't plan for a planet that kills people with air and soil that can't decide what it is."

"Your daughter."

Polk's jaw tightened. Something behind his eyes shifted—not his expression, his calculation.

"Haven't seen her in three years," he said. "Couldn't afford to fight for custody after the creditors took everything. She's twelve now, probably. Twelve and she remembers me as the guy who couldn't pay. If I die

on this planet, she gets a ghost. If I get home—even indentured, even broke—at least I'm standing there. At least she sees me and knows I tried to come back."

Not ideology. Parenting.

Ren understood that calculus better than she wanted to. She'd left behind a partner on Earth—not a child, but someone she'd been building something with before the debt spiral hit and she took the charter because staying meant becoming someone's dependent. The reaching-for-freedom narrative never accounted for the fact that sometimes freedom was the wrong word for running away.

"Garrett. If you signal and they come and you pay the extraction debt, you'll have less capacity to fight for custody than you do now. You'll be indentured. Worth less to her, not more."

"Maybe. Or maybe I get home and we figure it out."

"We don't have bandwidth for maybe."

"You don't," Polk said. "I do. Because my daughter is the math. Your spreadsheets are just the pencil."

The argument was escalating and going nowhere. Ren could feel it—the same structural inevitability as a wall taking on more load than it could hold. Polk wasn't going to be reasoned out of this because reason wasn't the foundation. His daughter was. And you couldn't engineer around someone's kid.

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She brought it to Cole that night.

He was in the main shelter reviewing equipment inventories by the rationed glow panels. Looked up when she entered, already reading her posture as bad news.

"Garrett's building a signal array," she said. "Northwest perimeter. He's going to try to call Earth."

Cole set down the tablet. "When?"

"Weeks. Four to six, depending on whether anyone actively helps him or just passively doesn't stop him."

He nodded. Not thinking. Calculating.

"If he signals, the contract voids. Asterlyne retrieves and extracts a penalty for early termination. Clause 14.3. Estimated cost: ninety-five thousand credits per colonist. At forty-four, that's four-point-two million. They won't bill the colony—they'll bill the signal initiator. Garrett personally. At standard compound rates, his granddaughter won't pay that off."

"He knows," Ren said. "He's decided his daughter is worth it."

"His daughter isn't here. Forty-four people are. His signal voids clause 9.4.2 for everyone. One man decides for all of us."

No heat in it. Just facts. Which made it worse—the cold certainty that his math took precedence over Garrett's love because numbers were reliable and love wasn't.

"You can't just tell people they're trapped," Ren said.

"I'm telling them they're free," Cole said. "They just have to survive it."

Ren heard the logic. She couldn't argue with it. The numbers were true. The clause was real. Signaling meant financial ruin, not rescue. Everything Cole said was correct.

She also heard what Cole couldn't hear in his own voice—that freedom defined by one man's arithmetic wasn't freedom. It was a different kind of contract. Survive on my terms or lose everything. Polk had heard

that exact math before, from creditors and courts and systems that told him the numbers were neutral while the numbers took his daughter away. Cole was running the same equation and calling it salvation.

"So what do we do?" she asked.

"I'll explain the financial consequences. Give him the numbers."

"He doesn't care about the numbers. That's not why he's building it."

"Then he shouldn't be building it. Whether he cares or not, the numbers are still true. His extraction bill will still be four-point-two million. His daughter will still inherit the debt—that's how generational obligation works. If he loves his daughter, he doesn't signal."

"You're going to tell a man who lost custody because he couldn't afford a lawyer that the math says he should stay trapped on another planet for his daughter's sake." Ren let that sit. "You're going to use his daughter as the reason he can't go home to his daughter."

Cole didn't flinch. "The math doesn't require him to like it."

"No. It just requires the rest of us to live with it."

Something passed between them. Not agreement. Recognition that they were standing on opposite sides of the same problem.

"What do you want me to do?" Cole asked.

"Something that isn't an equation. Something that acknowledges why he wants to signal without pretending it's unreasonable. Something that doesn't make him choose between his daughter and forty-four strangers."

"There is no option like that. Those are the actual variables."

"Then we need to change the variables. Give him a seat. A voice. Let him be part of the decisions instead of subject to them. He's not going to stop being afraid for his daughter. But maybe he stops building the array if he's not building it alone."

Cole looked at her with an expression she couldn't parse—something that might have been agreement or might have been the face of a man realizing the person across from him was solving a problem he hadn't known was the problem.

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Cole talked to Garrett anyway. Gave him the numbers. Ren didn't hear the conversation, but she heard the result—Garrett refused, he was finishing the array, debt calculations didn't change his mind. Cole had offered Garrett the math and Garrett had heard exactly what he'd heard from every creditor and courtroom that had separated him from his daughter: someone with a spreadsheet telling him what his love was worth and finding it insufficient.

What changed his mind was the scaffolding.

Four days after Cole's visit, two colonists from the shelter reinforcement crew—a woman named Lin and a man named Thomas who'd been quiet since landing, the kind of person who showed up for shifts and ate alone—were reinforcing the western wall of the secondary storage shelter. The crew was short-handed. Three people assigned to the rotation had drifted into Polk's orbit and weren't showing up for communal work anymore. Cole had adjusted the schedule to compensate, spreading the remaining crew thinner across the same number of tasks.

Lin and Thomas were operating a scaffolding rig they'd improvised from pre-fab framing and cable. Standard procedure required three people—one to brace, one to lift, one to secure. They had two. Thomas

was bracing and lifting simultaneously, doing the work of exhaustion and obligation in equal measure. Lin was overhead, securing a panel that weighed more than the two-person spec allowed.

The cable slipped. Not dramatically—no snap, no explosion of violence. A slow, grinding give. The kind of failure that happened when tension exceeded tolerance by a fraction and the fraction was enough. The scaffolding shifted. Lin fell. Thomas tried to catch the rig and the torque pulled him off the platform.

Ren heard the sound from across the settlement—compressed, metal and wood and something softer underneath. She ran. By the time she got there, Haruki was already kneeling.

Lin died on impact. Thomas died that night. Haruki sat with him through it, doing what he could, which wasn't enough because the injuries were structural and there were no surgical facilities on Parcel 16b.

Forty-two.

The deaths weren't the planet. The deaths were a construction accident caused by a crew that was too small doing work that was too heavy because the colony had fractured and the fracture had a body count now. The split—the quiet redistribution of trust away from the center, people drifting into factions because they didn't believe in the system—that was the load shift Ren had felt at month three. Lin and Thomas were the crack.

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She went to Polk. Not to Cole. To Polk.

He was at the array. Still building. He'd heard about Lin and Thomas. He didn't look defiant anymore. He looked like a man standing next to something he'd built while two people died somewhere he should have been.

"Two dead," Ren said. "Crew was short because your people weren't on rotation."

Polk didn't argue.

"I'm not blaming you. I'm telling you what the fracture costs. Not Cole's numbers. Bodies."

Polk looked at the half-assembled signal array—the copper wire, the salvaged antenna, the wooden frame he'd built with hands that knew construction the way Ren knew load-bearing walls. Hands that had built a co-op on Earth with the same conviction and watched it dissolve into debt.

"My daughter," he said. Quietly.

"I know. The council doesn't fix that. Nothing here fixes that." Ren kept her voice level. "But here's the offer. You dismantle the array. You get a seat on a colony council—elected, rotating, real authority over resource allocation and work assignments. Not Cole deciding alone. Not you deciding alone. A system that belongs to everyone who's still breathing."

"And the signal."

"Gone. Because you're right that one person shouldn't make decisions for forty-two people. Cole's right that the signal kills us all financially. Both things are true. The council is the only structure that holds both."

Polk was quiet for a long time. Ren watched his face—not calculating this time. Grieving. The apparatus he'd built out of wire and desperation and the ghost of his daughter's face, and it hadn't transmitted a single thing, and two people were dead, and the math still said he couldn't go home.

"If we survive long enough to invoke the clause," Ren said, "you own land. You own something that isn't debt. You become someone who built something instead of someone who owed something. Maybe that matters more to a twelve-year-old than a father who came home broke and indentured."

She didn't know if it was true. She said it because the alternative was watching the colony bleed out from a wound it was inflicting on itself.

Polk dismantled the array the next day. Didn't forgive Cole. Didn't trust the council. But he took the pieces apart with his own hands and returned the copper wire and the antenna components to the communal supply, and something shifted. Not unity. Not agreement. A framework for staying alive that wasn't built on one person's arithmetic.

It wasn't democracy. It wasn't dictatorship. It was the closest thing to governance that forty-two people on an alien planet could build out of grief and copper wire and the stubborn refusal to let the fracture kill them faster than the planet already was.

Forty-two alive. Fifty-four months left. The math was still brutal. But it was math they'd chosen together, and Ren wanted to believe that choice meant something, even if Cole couldn't quantify it.

Even if she couldn't quite make it mean enough.

# Chapter 08 — "Roots"

Month five took two of them.

The first was a man named Osei. Respiratory infection. Not the spore cascade that killed Lev, but something quieter. A cough that started dry and deepened over four days until it sounded structural. Haruki treated it aggressively. Antibiotics, fluid intake, rest protocols. Osei improved on day three. Then he didn't. His lungs filled with fluid that shouldn't have been there. Something in the planet's atmospheric chemistry interacted with the infection in ways Haruki couldn't model because he didn't have a model for this biology.

Osei died on day seven. Haruki's hands were still when he came to report it.

Forty-one.

The second was exposure. A colonist named Farin on perimeter check during a temperature swing — one of the sudden drops Dana had been tracking but couldn't yet predict at this magnitude. Farin was dressed for the afternoon's warmth. By the time the two-hour check-in was missed and the search party went out, hypothermia had done its work. They found her two hundred meters past the eastern marker, sitting against a spiral tree like she'd decided to rest.

Forty.

Cole adjusted. He always adjusted. Updated the respiratory screening protocol for anyone working in spore-proximity zones. Required thermal gear for all perimeter shifts regardless of afternoon temperature. Tightened buddy-system enforcement. The protocols were sound. The kind of systems design that would have earned a commendation in his logistics career on Earth.

The protocols didn't bring anyone back.

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Dana Wen was in the soil lab — a reinforced tent with a raised analysis table and equipment they'd salvaged or improvised from mining-grade supplies. Cole watched her from the entrance. She didn't know he was there. He'd learned to move through the settlement without announcing himself. Observation without intrusion.

She was covered in the lavender dust of Parcel 16b's soil. In her hair, under her fingernails, on the cuffs of her shirt where she'd wiped her hands all day. Methodical. Taking samples from marked collection points, running them through a centrifuge they'd powered by hand that morning, noting something on a paper tablet without looking up, then moving to the next station like she was the only person in the colony who understood what work actually was.

Cole had reviewed her progress reports. This month's read like the notes of someone assembling a map. Not random data points. A pattern.

"You've found something," he said.

Dana looked up. Didn't startle. She had good control of her nervous system.

"Readings from early settlement through now," she said. She set down the tablet and gestured to the analysis table. "I've been tracking mineral content, bacterial colonies, pH variance. Cross-referenced against the planetary rotation data from the original survey pack."

Cole moved closer. The tablet showed a graph — mineral concentrations plotted against time. Noise in the data, but underneath the noise was a rhythm. Clear as pulse.

"It's not random," Cole said.

"It's not random," Dana confirmed. "It follows the orbital position. When the planet's closest to the sun, these minerals are accessible. The bacterial colonies are more active. When we move out of alignment, everything shifts dormant. It's a cycle. Maybe seasonal variation based on thermal cycles we can't directly measure yet, maybe something more complex. But it's predictable."

She said this the way she said everything. No inflation. No self-congratulation. Data delivered like a work order. Part of why Cole believed her.

"I have six native species that track with the high-activity windows," Dana said. "We can plan what grows when. Stop losing crops to random soil shifts. First time we know what to plant and when to plant it."

Cole ran the math. Predictable growing windows meant extended rationing ceilings. More reliable yields. More margin. They were at forty. The trajectory had been shallow — barely holding, always at the edge. Dana's map changed it.

"How confident are you?"

"Sixty percent. Seventy if we get another two months of data. The pattern holds so far but the planet's still teaching us things. I'm not going to promise more than I can deliver."

"Show me."

She walked him through it. Ninety minutes. The bacterial analysis, the mineral mapping, the way she'd isolated variables by testing against samples from different points across the settlement's history. The work was elegant. Not because Dana was trying for elegance but because she was trying to solve a problem and precision was the byproduct.

By the end, Cole was running new projections. If the cycle held, they could adjust crop schedules. If they adjusted schedules, rationing could loosen. If rationing loosened, people ate better. Infection rates dropped. Strength levels rose. Work output increased. The math cascaded in the right direction for once.

"Present this to the council tonight," Cole said. "Full briefing. Let them ask questions."

"I'd rather you did. I'm a scientist, not a politician."

"I'll do the presentation. You sit at the front. Answer the technical questions. They need to see where the certainty lives."

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That night, the colony ate a full meal for the first time since landing. Not rationed. Not stretched. Full portions. The council had authorized it — celebrating Dana's breakthrough, the opening of a timeline that didn't end in slow starvation.

People smiled. Cole watched it happen from the edge of the gathering. Joss Carver laughing at something Haruki said. Garrett Polk actually eating instead of performing nutrition. Ren making eye contact with people instead of studying the ground. The shared meal did something the council votes couldn't. It reminded them they were a group instead of a collection of individual survival calculations.

Cole ate. Didn't celebrate.

He was thinking about Elena Vasik.

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He opened the data core that night. First time since month two, when Elena's entries had stopped at the boundary of what Cole could absorb. Nine survivors at the end of her second month. He'd closed the file and rationed the rest for later.

Later was now.

Elena's Month Three logs read like a mirror held at Cole's own settlement. Social fracture. Two of her nine had started operating independently — hoarding supplies, skipping communal meals, building a separate water cache. She documented the split the way she documented soil pH. As a variable she was monitoring without intervention. Her sentences were getting shorter.

Month Four. Another death. Medical complication from a wound that should have healed. Elena noted the cause, adjusted the rotation, added the name to a list growing in only one direction.

Month Five. Elena's colony found something in the soil. Not the full pattern — she was working without Dana's instrumentation — but the outline of it. The beginning of understanding that the planet operated on a cycle, that the chemistry wasn't random. Her entries picked up. Longer sentences. Questions driven by curiosity instead of desperation.

Month Six: "The soil follows the orbit. We can predict the growing windows. We can plant to the cycle instead of against it."

She'd found it. The same breakthrough Dana had presented to the council three hours ago. Seven years earlier. Fewer people.

Months Seven through Eleven: stabilization. Elena's colony found its footing. The survivors held. No deaths. Crop yields improved. The social fracture didn't heal, but it stopped widening. Elena's entries settled into something resembling routine. Weather observations. Soil data. Harvest projections. The controlled cadence of a scientist who believed the pattern would hold.

Her final Year One entry was five words.

"We're going to make it."

Cole closed the data core. Sat in his shelter. Felt the sentence settle against his chest like a stone placed carefully on a grave.

Elena had been right about the soil. Right about the cycle. Right about the growing windows. Every piece of data matched what Dana was finding now. The science was sound.

The clearing twelve kilometers west was what sound science looked like at the end.

He didn't open the Year Two files. Whatever lived in those entries — whatever turned five words of certainty into collapsed shelters and a sealed data core — he'd read it when the math demanded it. He was rationing Elena the way he rationed everything else. Taking what he could use. Saving the rest.

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Three days after the full meal, Haruki appeared in Cole's doorway.

"We have a situation." Level voice. Which meant serious.

A colonist named Sarah. Construction crew. She'd reported to Haruki with a cut on her forearm — shelter repair work, minor. Cole had pulled her from the construction rotation the same day. Standard protocol. Light duty until Haruki cleared her. He'd redistributed her tasks across the remaining crew, adjusted the schedule. The system worked the way it was supposed to.

The wound had been cleaned, bandaged, treated with standard protocols. Bandage changed daily. Minor inflammation yesterday. Nothing alarming.

This morning the inflammation had progressed. Infection setting in fast. Haruki had escalated. Broad-spectrum antibiotics, wound cleaning, rest, fluids. The textbooks said this should respond within forty-eight hours.

"She's getting worse," Haruki said. "The antibiotics aren't touching it."

Cole went to the medical shelter. Sarah was on an examination bed, feverish and confused. Her arm swollen from the elbow down, the infection advancing — not localized the way human infections progressed but colonial. Something establishing an outpost and expanding.

"It's the planet," Haruki said. "The wound was clean. The pathology is — I don't have a framework for this."

The antibiotics were designed for Earth infections. This wasn't an Earth infection.

Sarah died on Day Four of the fever. Haruki was with her. Cole was in his shelter, adjusting the roster. Removing her from assignments, rotation schedules, the rationing tier. He'd memorized all the ways to make a person into a blank space in a system.

Thirty-nine.

He'd done everything right. Pulled her from rotation the hour she reported the injury. Adjusted the schedule. Haruki had treated the wound within minutes, followed every protocol, escalated at the first sign of resistance. Every input correct. Every response timely. Every optimization in place.

The planet's biology was just different. The variables underneath Cole's model were alive, evolving, indifferent to his calculations. He'd built a system for Earth and deployed it on a world that hadn't agreed to be optimized.

Sarah was the proof. Not that Cole had failed. That the math itself operated on assumptions the planet didn't share.

He thought about Elena's Year One ending. "We're going to make it." A woman who'd cracked the code, mapped the cycle, stabilized her colony. All of it correct. None of it enough.

Thirty-nine alive. Dana's map held. The soil cycle was real. The colony would eat.

The planet would keep teaching them rules one at a time. Usually by breaking something they thought was safe.

# Chapter 09 — "The Count"

Six months. Cole stood in the settlement's makeshift administrative shelter—a reinforced pre-fab with a table that held the colony's maps, equipment manifests, work rotation schedules, and the running tally of who was alive and what they could do with their bodies and skills. He'd been coming here at dusk most days, reviewing the day's data, adjusting projections, doing the invisible work that made it seem like the settlement ran on something other than mathematics and fragile luck.

Thirty-nine.

He'd started with forty-seven. Eight gone in six months. Spores, structural failures, a scaffolding collapse during the faction split, exposure, infections the antibiotics couldn't reach. The specifics varied. The direction didn't. Eight subtracted from forty-seven left thirty-nine, and thirty-nine had held for three weeks.

But holding was the right word. The agricultural cycle was stable—Dana's breakthrough had loosened the rationing enough that people weren't eating at a permanent deficit. Water collection was running on schedule, the filtration system Ren had redesigned was holding, and the shelters had been patched enough to survive the UV warping that ate through the original composite. Fragile. Functional. Ren's council system was working—or not working, but functioning in the way that systems do when everyone involved has agreed to pretend the framework matters. Garrett Polk had been neutralized, which was a word Cole used in his head without examining it too closely. Polk had wanted to signal for rescue. Ren had offered him a place on the council—real input, real voice in the decisions that affected the colony. In exchange, Polk dismantled the signal array. Not because he'd changed his mind about the situation, but because the promise of agency was enough to make him believe he had options again.

It wasn't democracy. Cole knew that. Democracy required actual equality. What they had was informed consent to a hierarchy that wore the shape of shared decision-making. But the colonists had accepted it. Some because they believed in it. Some because the alternative was isolation. Most because the settlement had enough work to keep everyone distracted.

Cole closed the day's reports. The shelter was dimming as the sun moved toward the horizon. He had maybe twenty minutes of useful light left. Enough time to eat something or do something useful. He chose neither. He chose to sit at the table with a piece of paper and the knowledge that he owed a message to a daughter who didn't know she'd been abandoned on an alien world.

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The message had to be simple. Cole understood that much. Lily was nine. Nine-year-olds had attention spans measured in paragraphs, not pages. He'd tried to write something the day after landing and stopped after the first sentence. Everything that came after felt like a lie or an admission, depending on how he tilted the words.

He started with the most honest thing he could think of:

*Lily,*

*I'm on a planet very far away. The trees here are unlike anything on Earth. They have spiral trunks that twist as they grow, and they hum at a frequency you feel more than hear. At night, when the wind moves through them, the sound reminds me of—*

He stopped. Reminds him of what? Her mother singing? A lullaby he'd stopped playing years ago? He scratched through the line.

*The trees here hum. The sound is below human hearing, but you can feel it in your chest if you stand under them long enough. It's beautiful and strange and I think you'd want to draw it. You like drawing things that*

*don't exist on Earth. You draw horses with wings. You draw castles that float. This planet is full of things that would be impossible in your sketchbook, except they exist, so maybe the sketchbook was the universe's way of preparing you.*

He stopped again. Too much. Too trying. Cole knew how to talk in his own head. Talking to Lily was different. She was real and distant at the same time. More real for the distance.

He started over.

*Lily,*

*I'm on a planet very far away. The trees are beautiful. I wish you could see them.*

That was true. That was the most useful truth he had. He folded the message and set it aside. Three sentences. It had taken him two hours to arrange them so they didn't sound like a man saying goodbye.

He didn't know when the message would reach Earth. The relay satellites had their own orbital mechanics. The ship that would eventually carry it back had its own schedule. Months, probably. Maybe longer. Lily would read it whenever it arrived. By then she'd be older. The trees would still be here. Cole might not be.

He pressed his thumb against his eye and held it there until he stopped thinking about any of that.

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The forest changed after dark. The hum deepened—not louder but lower, finding frequencies Cole hadn't heard in six months on this planet. It pressed against his sternum like a hand. He stood outside his shelter and listened to something he couldn't name shift underneath the night.

Then the soil temperature alert Dana had rigged two months ago went off. Six degrees in one hour. Not the air. The ground.

Dana was at his shelter before the alert finished sounding. She'd been monitoring the overnight readings. Of course she had.

"Soil samples," she said. "Now."

They moved to the lab without talking. Dana went straight to the analysis equipment—no preamble, no exhaustion. Problem reduction. Data collection. The specific focus that Cole recognized as her truest mode.

The samples took thirty minutes. Cole stood at the edge of the lab and watched numbers populate her screen. Dana's patterns were still there—the predictable cycles she'd mapped—but underneath them was a new variable. Temperature shift. Mineral composition trending in a direction she hadn't tracked. Bacterial colonies that should have been dormant beginning to activate.

Dana's face when she turned from the screen. Cole had seen her process bad data before—conflicting numbers, ambiguous results. This wasn't that. This was recognition.

"It's starting," she said. "The second cycle."

The words hit something that was already waiting. Elena Vasik had written the same phrase in her Month Nine entry—Year One, the logs Cole had finished three days ago. *The second cycle.* Elena had named it. Had seen it arrive. What she did about it, how her colony responded, whether the second cycle was what finally killed them—that lived in the Year Two files. The ones he hadn't opened.

"How long?" he asked.

"Two to three weeks for the shift to complete. After that, we're in new territory."

Thirty-nine people. Systems built for conditions that were about to shift underneath them. Everything they'd constructed in six months tuned to a planet that had just changed its pitch.

"We need to go to the council," Dana said. "Tonight."

"I know." Cole's voice was flat. Not anger. Arithmetic. Six months of optimization undone by an orbital position.

"We tell them everything," Dana said. "The data. The timeline. No managing it."

Cole looked at her. Soil on her cuffs, under her nails, in the creases of her knuckles. She hadn't been entirely clean since Month Two.

"Council tonight," he said. "You present the data. I'll do the rest."

Dana gathered her samples. Cole stayed in the lab as the darkness outside became complete. The hum pressed through the shelter walls—the planet leaning against the structure, not hostile, not welcoming. Just turning the way planets do. Indifferent to what had been built on the surface.

Six months. Thirty-nine alive. Fifty-four months to go.

He walked back to his shelter. Elena Vasik's data core sat on the shelf where he'd left it after finishing Year One. The file that ended with five words: *We're going to make it.*

Year Two was the rest of the story. The part where certainty met the planet and lost.

Cole picked up the data core. Held it in the dark.

He opened the file.

# Chapter 10 — "Second Cycle"

The shift happened overnight. Not literally—Cole had watched it coming for two days through Dana's soil samples, tracking the temperature curve and mineral fluctuation as the planet's orbital mechanics tipped toward the second cycle. But once it began, it moved fast enough to feel like overnight.

The vegetation died first. Not dramatically. Just stopped. The Earth crops that had been showing recovery under Dana's careful management—the wheat variants, the bean strains, the root vegetables they'd been nursing along—turned brown in their beds like someone had flipped a switch. Not disease. Not pest damage. Just the signal that their usefulness had ended and the planet was moving on.

Cole was checking the north agricultural plot on Day Seven of the shift when he first saw it with full clarity. The same plots that had fed the colony's optimism three days ago were now showing widespread dormancy. The lavender soil had darkened—more minerals, denser composition. The light from the planet's sun looked different too, shifted another degree toward green. It wasn't visible in the way weather changed on Earth. It was visible the way a mood change is visible if you're paying attention to a person's face: a small shift that meant everything underneath had reorganized.

Dana was there with him, pulling samples with mechanical precision, not commenting on what they both understood perfectly. The advantage the colony had thought they'd gained was evaporating.

"Three weeks," Dana said. She was reading her tablet, cross-referencing the new soil data against her cycle projections. "Maybe four. This matches the pattern from the data. The high-activity period is ending. The planet is entering dormancy phase."

"How dormant?" Cole asked.

"Complete," Dana said. "No Earth crops will survive the dormancy window. The soil composition becomes actively hostile to the microbiology they depend on. Some native flora might persist but it cycles down—becomes mostly inert. If your species aren't designed for the dormancy phase, the planet stops feeding them."

Cole did the arithmetic in his head. The colony was at thirty-nine. Dana's breakthrough had bought them confidence, but it had bought them seven weeks of actual extension. They'd built assumptions on top of that extension, increased consumption, prepared for expansion. All of that math was now being subtracted.

"Elena's logs?" Ren asked. She'd arrived while Cole and Dana were examining the plots, and Cole hadn't noticed her approach. She was learning to move quietly through the settlement the way Cole did, though her reasons were different. Cole was trying to observe without influencing. Ren was trying to listen without being asked to contribute.

"Month nine for Elena," Cole said. "She started serious starvation at this point. Her colony was smaller—she could stretch resources. We have more people and the same amount of dormancy."

He didn't need to finish the calculation. Ren was capable of running it herself.

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Cole read Elena's Year Two entries that night. The administrative shelter with the door sealed and the hum pressing low through the walls.

Elena's Year Two opened at her Month 13. The clinical precision of her earlier entries was gone. Not replaced by desperation — not yet. By a kind of honesty that arrives when the thing you've been measuring starts measuring you back.

*Month 13: Second cycle has deepened. All cultivated species dormant or toxic. Native edibles reduced to three viable candidates. Yield insufficient for nine. Rationing at levels that will produce cognitive decline within six weeks.*

Nine people. Cole had thirty-nine. Four times the population, same soil, same orbital mechanics. Elena had found the cycle too. Her data had been real, her breakthrough sound. The planet didn't care about the data.

*Month 14: Two members have resumed independent food caching. The colony's social contract is under revision — not formally, not through conversation, but through the slow withdrawal of shared assumption. People eat alone now.*

Cole recognized it. The same thing he'd already started seeing in his own settlement — people carrying themselves differently, spending time in shelters instead of common areas, conversations getting shorter. Elena had watched it happen and recorded it with a scientist's precision. Recording wasn't preventing.

*Month 15: A biological mat has appeared at the forest edge. Details in attached data. It advances slowly. We have time. I think we have time.*

He marked the entry and stopped reading. The rest could wait. He was rationing Elena the way he rationed everything else — taking what he could use, saving the rest for when the math demanded it.

But the parallel was exact enough to sit in his chest. Elena's advantage had been scale — nine people could decide faster, consume less, relocate without councils or votes. Her disadvantage was the same: nine people had less labor, less redundancy, less capacity to absorb loss. One person stopping cooperation wasn't a percentage problem. It was an eleventh of everything.

Elena had the data. She had the pattern. She had everything Dana had given Cole, minus the people to make it work.

Cole had thirty-nine people and Dana's soil map and six months of agricultural data Elena never had. Those were his advantages. His disadvantage was thirty-nine mouths instead of nine, and a dormancy window that didn't care about headcount.

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The watchers in the tree line became more active. Cole had noticed them before—shadows at the perimeter, movement in the canopy, the sense of being observed that was real, not paranoia. In the first cycle, they'd kept distance. Now they moved closer. Bolder. One morning, a creature that looked like it was built from cervid and praying mantis components walked through the outer perimeter and examined a supply crate before wandering back into the forest. It wasn't aggressive. It was curious.

"They're hungry too," Garrett Polk said at a council meeting. He was pointing at the perimeter watch logs, at the increasing frequency of fauna sightings. "The whole planet shifts cycles and everything that lives here needs to adjust. We're not special. We're just one more thing that needs feeding."

Cole didn't argue. Polk was right. The colony wasn't being hunted. They were just living in a system that was radically indifferent to whether they continued or stopped. The shift in fauna behavior wasn't targeting. It was adaptation. The planet was teaching every organism at once: the rules have changed. Adjust or die.

The soil rejected Earth crops entirely by Week Three of the dormancy phase. Dana had anticipated it—her maps were holding—but anticipation didn't change the fact of standing in the agricultural zone and watching months of careful cultivation become useless in a matter of days. The colony's food security, which had felt solid one month ago, suddenly looked like a calculation built on sand.

Cole issued the new rationing protocols. Reduced portion sizes. Adjusted work rotations to account for lower caloric intake. Shifted labor focus to identify and cultivate native species that Dana was mapping as dormancy-phase viable. The mathematics were clean. The implementation was brutal.

Hoarding started on Week Two of the rationing adjustment. Cole found it the way he'd found most problems—not through inspection but through observation. People moving differently, carrying weight differently, spending time in their shelters instead of the common areas. Cache sites discovered: a colonist maintaining a private food store under her shelter, another hoarding water containers in a locked compartment of the equipment shed.

He reported both to Ren.

"They're afraid," Ren said.

"Yes," Cole said.

"So they hoard. Makes sense. Fear does that."

"It breaks the colony," Cole said. "If food distribution becomes unpredictable, if people start hiding supplies instead of pooling resources, the morale failure cascades. We've accounted for voluntary compliance. If people start acting against the collective, the margin disappears."

"So what do you want to do?" Ren was watching him with an expression he was learning to read as something between exasperation and respect. She understood what he was trying to solve. She didn't agree with the solution he was reaching for. "Punish them? Make an example? Force compliance through fear of something worse than starvation?"

"I'm asking what you recommend," Cole said.

"Transparency," Ren said immediately. "Tell everyone exactly how much food exists. Tell them the timeline. Tell them the dormancy window is going to be hard and they need to understand the math so they can trust the system."

"That information is destabilizing," Cole said. "People hear 'critical shortage' and it triggers panic buying, not rational cooperation."

"People hear 'secret shortage' and they prepare for betrayal," Ren said. "At least if you're transparent, they can see the choice being made. They can participate instead of watching you calculate in private and guessing whether they're already dead."

Cole understood what she was actually saying. She was saying: stop treating the colony like a workforce you manage. Treat them like a group that gets to know what's happening.

He called for a full colony assembly. Didn't announce it formally—he had learned that the most effective way to gather people was to let them find out that the meeting was happening. Word spread fast in a settlement of thirty-nine.

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He stood in front of them with the inventory data and the timeline projection and didn't soften any of it.

"We have seven weeks of rationed food if the dormancy phase holds as Dana's projections indicate," Cole said. "The dormancy window lasts approximately sixteen weeks. The gap is nine weeks. During those nine weeks, we'll need to have identified and cultivated enough native species to bridge the shortfall. Dana has narrowed it down to approximately thirty candidates that remain viable during dormancy. We'll test them for edibility and yield. Some will work. Some won't."

He showed them the rationing tiers. Showed them the work rotation that would be required to keep searching for alternate sources. Showed them the margin—what remained of it.

"We'll be hungry," Cole said. "Significantly. But the calculation shows we can survive the dormancy window if the colony acts as a unit instead of as individuals trying to protect themselves."

The silence after that announcement was profound. Not angry. Not defeated. Just the sound of thirty-nine people recalibrating their understanding of what they were attempting.

"Why tell us?" someone asked. Joss Carver, his nervous energy focused into genuine curiosity instead of anxiety. "I mean—I'm glad. I just mean, wouldn't it be easier to manage us if we didn't know?"

Cole thought about lying. It would have been simpler. A logistics problem solved through information control.

"Because you're not a system I manage," Cole said. "You're a colony I'm part of. The system works if we're deciding to stay in it together. If I'm running calculations on you instead of with you, the system fails." He paused. "There was a colony here before us. Fourteen people. Their leader had the same data we have. Same soil cycle, same projections. She wrote 'We're going to make it' at the end of her first year. By the second year, her people had stopped eating together. Then they stopped talking. Then there were six."

Ren was watching him with an expression that looked almost like surprise. Or maybe recognition. A small shift in how she was reading something about him.

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The first three weeks of dormancy were the hardest. Rationing meant visible hunger. Visible hunger meant visible fraying. Conflicts over work assignments became sharper. Criticism of resource allocation became louder. The colony was hungry and afraid and aware that both conditions were deliberate mathematical choices made by the person standing in front of them.

Cole sat through it. Took the criticism. Adjusted where the calculations showed adjustment was needed, held firm where the math didn't allow flex.

Dana spent those three weeks essentially living in the forest. Out at dawn with Haruki's medical oversight, gathering samples, testing specimens for toxicity and nutrient density. Testing them on herself sometimes. Cole watched her do it once and didn't tell her to stop.

By the end of the third week, she'd identified three native species that would remain edible through the dormancy window and could be cultivated in sufficient quantity to bridge most of the shortfall.

She announced it at a colony gathering, standing next to Cole the way they'd done for the agricultural announcement in the last cycle. But this time, the tone was different. Not celebration. Survival.

"The first is a root-like structure in the spiral trees," Dana said. "High in starches and some proteins. Not palatable until processed. The second is a seeded fruit that ripens during dormancy—the timing is opposite to the Earth plants. The third is a fungal compound that my preliminary analysis shows can support bacterial colonies we use for fermentation and nutrition synthesis."

She was reading from notes. Her voice carried the specific exhaustion of twenty days in the forest and coming back with just enough.

"We'll need to shift agricultural focus immediately," Dana continued. "Stop trying to keep Earth crops viable. Concentrate on cultivating the dormancy species. The yield will be lower than the cycle-one projections, but it will be sufficient."

Cole watched the colony process that information. The hunger wouldn't end. But it wouldn't kill them either. There was a difference between those two facts, and Ren had been right to insist on transparency. The colony could see the path, see the work, see the weight. They could choose to walk it.

They wouldn't starve. They'd just be hungry for a long time. Cole could work with that.

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Cole was reviewing the new agricultural rotation with Ren when he realized the planet had changed the color of the sky another degree. The green was thickening. The light was becoming something his eyes kept

trying to correct for, like looking through water that wasn't there.

"How many colonies do you think Asterlyne has sent here?" Ren asked.

"We know of one before us," Cole said. "Elena's. There were probably others. Probably failures they didn't document well enough to leave behind."

"Do you think anyone on Earth understands what this planet does?" Ren was sitting with her back against the wall of the equipment shelter, watching the agricultural zone where colonists were already beginning to transplant the dormancy species. "Like, actually understands? Or do they just look at profit margins and casualty projections and decide it's worth the cost?"

Cole thought about Asterlyne's data package, about the redacted appendix, about the viability score inflated by twelve points. About the charter designed to be unread and the clause that had only survived because no one expected anyone to live long enough to invoke it.

"They understand the math," Cole said. "They don't understand the people it kills."

Ren didn't answer. She was watching two colonists working the transplant rows in the shifted green light without complaint. People adjusting to new rules because someone had told them what the rules were.

"Your council held through this," Cole said.

"It's not mine," Ren said. "That's why it held."

Cole went back to the rotation schedules. The colony would eat less and work more and hold together because thirty-nine people had decided to hold together. Not because the math demanded it. Because Ren had been right: they'd chosen to see the weight, and they'd chosen to carry it.

The planet had turned a page. The colony was still reading.

# Chapter 11 — Creep

Something was wrong at the forest edge.

Cole noticed it during dawn inventory on Month 8, Day 3 — a gray-brown smear in the lavender understory, twenty meters beyond the perimeter markers. A discoloration that hadn't been there the previous evening. The soil around the settlement had been shifting all month, temperatures dropping, drainage patterns changing, the whole planet recalibrating for its second pass through the orbital cycle. A new discoloration was one more variable in a system already too full of them. He logged it. Moved on.

By Month 8, Day 7, the smear had expanded to three meters.

By Day 10, it was a band ten meters wide and advancing at a rate Cole could measure against marker stones. Slow. Patient. A centimeter an hour, maybe less. He'd set Ren to charting it — time-marked photographs every four hours, distance measurements, soil samples from the leading edge. The thing moved like erosion, like time made visible.

It was Dana who named it. "The creep," she called it in the weekly resource meeting, and the word stuck because it was accurate. It crept. That was all it did — moved forward with no apparent urgency, breaking down everything organic in its path.

Haruki collected samples. Ran them past his makeshift field lab. Came back with observations: fungal network of some kind, but nothing matching Earth taxonomy. Mineralized base structure — active, flowing, consuming. It dissolved plant matter on contact, the way mold breaks down a corpse, except faster, more complete.

Cole recognized it before Dana said the word. Elena's Month 15 entry, the last one he'd read in her Year Two logs, the one that had sat in his chest for weeks: *"A biological mat has appeared at the forest edge."* He'd read that line and set the data core aside, rationing Elena the way he rationed everything else. Now the mat was here. Not in Elena's logs. On his perimeter.

He went back to the Year One entries he'd already read months ago. Month 9 landed differently now: *"The fungal bloom has accelerated since we began tracking it. Initial theory: seasonal response to orbital position. The structure maintains internal moisture and temperature regulation — it's not dying back. It's feeding."*

Month 11: *"We attempted containment with fire. The accelerant ignited, but the creep adapted — routed around the burn, moved deeper into the soil. The burned section appears to have accelerated its growth rate. Chemical barriers purchased from Asterlyne inventory: effective for 4-6 weeks. After that, adaptation. The thing learns."*

He opened Elena's Year Two files past where he'd stopped. Month 16: *"The creep has consumed the eastern shelter line. We're relocating again. Running out of room."*

Cole closed the data core and walked to the perimeter to watch it move.

It was beautiful in a way that made him uneasy. The texture was almost organic — a network of filaments so fine they looked like lace from a distance, except the lace was dissolving the soil beneath it, breaking down the biomass left by the second-cycle plant die-off. The color was gray-brown-lavender, shifting in the low light. It didn't move like a wave. It moved like breath. Expansion and rest, expansion and rest.

"It's not attacking," Dana said, coming up beside him. She had a specimen container in one hand, a measurement tape in the other. She'd spent three days collecting samples, running composition analysis on the compound structure. "It's decomposing."

"It's advancing on the settlement."

"It's doing what it's evolved to do. The second cycle brings dormancy, die-off, dead matter. The planet has a cleanup system." Dana crouched at the creep's edge, studying the gradient where the fungal network met unaffected soil. "If we were a forest, we'd welcome this. Nutrient cycling. Renewal."

"We're not a forest."

"No. But we're in one. There's a difference."

Cole understood the distinction intellectually. It didn't change the math. At current advancement rate, the creep would reach the perimeter shelters in seventy to eighty days. The core settlement — where the medical facility was, where the food stores were, where most of the colonists slept — was another two hundred meters inland. Call it eleven weeks. Call it the time until the second cycle deepened and nothing was predictable anymore.

He brought it to Ren.

"We need to relocate," she said. Not a question. A statement she'd already mapped the implications of.

"We need to understand it first. Elena tried containment. Fire didn't work. Chemicals bought them six weeks. After that—"

"After that, they abandoned the site and lost the agricultural work they'd already done. We don't have Elena's luxury. We're at thirty-nine. She started with fourteen."

Cole didn't correct her. She meant Elena was running the math differently — fewer people meant earlier desperation meant faster decisions. Ren's logic was sound. Elena's colony had mobility. The Free Colony was rooted.

Dana led them deeper into the research, setting up a makeshift lab in one of the cleared shelters. She collected samples from different points along the creep's boundary, analyzed the internal structure under magnification — a repurposed engineering microscope, never meant for biology — tracked the compound's response to various external stimuli. Ren helped. Joss Carver volunteered to assist, ferrying equipment, taking notes with the concentration of someone trying to prove his utility. Cole watched Dana work the way she always worked: hands steady, face neutral, the specimen container held like it mattered more than the thing consuming it. She'd handled Marcus Reeves's shelter samples the same day the body was found, pulling creep specimens from the wall breach while Haruki processed what was left inside. She hadn't hesitated. Cole filed that.

It took Dana two weeks to map the pattern.

The creep's composition shifted with depth. The surface layer was aggressive — breaking down plant matter at acceleration. The middle layers were transitional, the filaments more tightly woven. The deepest sections, several centimeters down, were crystalline. Mineralized. Dormant.

"It's storing," Dana said, showing Cole the cross-section. The compound was sealed in a transparent container, and under the magnification lens, the internal structure was almost geometric. "Look at the ratio of active to inert material. The active processes break down the biomass, and then the mineral structures preserve and store what it's extracted. It's not just decomposing. It's sequestering. Building something."

"Building what?"

"I don't know yet. But it's intelligent in the way a system can be intelligent. It's responding to the planet's needs, not expressing malice. It's not our enemy, Cole. It's the planet's immune system. It just happens to be cleaning a space we're occupying."

The distinction was fine enough that it didn't matter. The creep was still advancing on the settlement.

Polk wanted to burn it. Stood up in the council meeting and said they should trench the perimeter and light it. Cole vetoed. Polk pushed back — wanted to know why they were studying the thing instead of fighting it. Cole could have explained Elena's data, the fire adaptation, the chemical failure. He didn't. Polk hadn't earned the logs. Polk wanted action, immediate and decisive. Cole wanted a solution that would hold. Those two wants collided in the daily council meetings, where Polk's frustration mounted and Cole's silence only deepened it.

"You've got a plan," Polk said after one session, catching Cole outside the council shelter. "I can see you running numbers in your head while the rest of us talk. So either share the math or stop pretending this is a democracy."

Cole looked at him. "It's not a democracy. It's a survival problem."

"Same thing, where I come from. The people who do the math get to make the rules, and the rest of us get to live with it."

Polk walked away. Cole watched him go and thought about the signal array, the rescue frequency, the man who'd wanted off this planet from the moment he arrived. Polk's instincts weren't wrong. They were just expensive.

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On Month 9, Day 2, one of the perimeter shelters ceased reporting.

It was a routine check — the supply tech who worked the eastern shelter line didn't show for the morning count. Cole sent a search team. They found the shelter intact but unsealed. The opening in the wall had been gradual — the creep had grown through the composite material, patient and methodical, following the path of least resistance. The colonist — a man named Marcus Reeves, thirty-one, former software engineer who'd taken the settlement logistics role because it kept him occupied and away from processing why he'd left Earth in the first place — was still inside.

Haruki did the field examination. Cole watched from a respectful distance, close enough to understand, far enough to preserve something like dignity for what remained. The creep had been indiscriminate. Reeves's lower body was dissolved to the point where the fungal network had replaced what should have been there — not in a clinical, educational way, but in a way that made clear the organism considered organic matter as raw substrate. His upper body was intact. His face was untouched. His eyes were closed, or dissolved to where it appeared they were closed.

Haruki said, "He was unconscious before the significant dissolution." Nothing else. The clinical economy of it was the mercy — what Haruki chose not to speculate about, the questions he didn't answer.

Cole helped Haruki carry what was left to the processing area. They didn't call it cremation — the planet's high-temperature cycles and the shelter's heating system could serve the same purpose. Haruki prepared the body with the same precision he applied to everything, which was the only dignity available: careful, methodical, without performance or false ceremony.

No one saw the body except Cole and Haruki. Cole kept it that way, not through explicit instruction but through presence — staying near the processing shelter, making it clear that this particular space was closed. The colonists understood. They always did. Death was common enough that they'd learned the etiquette of it.

But the etiquette was different this time. The other deaths had been comprehensible — spores, falls, infections, structural failure. Human-scaled problems with human-scaled causes. The creep wasn't human-scaled. The colonists didn't know what had happened to Marcus Reeves's body, but they knew the shelter wall had been breached from the outside by something that moved slowly enough to be invisible and fast

enough to kill a man in his sleep. That knowledge sat in the settlement like a change in air pressure. People stopped sleeping near the perimeter. Two colonists asked to move their shelter assignments inland. Three asked Haruki how it happened and got the same clinical economy Cole had received. Nobody asked Cole. They'd learned that Cole's silence on the specifics of death meant the specifics were worse than the fact of it.

The count dropped to thirty-eight.

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That night, Cole spread Ren's time-marked photographs across the floor of his shelter and did the work.

The creep's advance wasn't uniform. The photographs showed it moving faster through areas with dense organic material — the composting zones, the plant waste from the agricultural experiments, the paths where foot traffic had compacted biomass into the soil. It moved slower over rocky patches, mineral-rich ground, areas where the planet's bedrock was close to the surface. He marked the rates on the map. Drew the projection lines. Checked them twice.

Eleven weeks to the core shelters at current speed. But the current speed was the second cycle's opening phase. Elena's logs said the creep accelerated as the cycle deepened. If it doubled — and Elena's data suggested it would — the timeline compressed to six weeks. Maybe five. The perimeter shelters were already compromised. The northern agricultural plots, where the dormancy-phase species Dana had cultivated were the colony's only reliable food source, sat directly in the creep's path. Seventy days. Maybe less.

Elena had been competent. She'd had the data, the scientific precision, the survival instinct. She'd relocated three times. The third relocation was made with six people remaining. The creep had consumed the other eight — not through malice, through patience. Elena's colony hadn't lost a war. They'd lost a race against something that never tired, never stopped, never needed to win quickly.

Elena never solved this. Cole sat with that fact the way he sat with all the facts that mattered — quietly, completely, until it stopped being information and became weight.

He pulled her logs again. Re-read the containment entries. Her conclusion, written in Month 17 after the third relocation: *"The creep isn't something we can fight. It's something we can only move ahead of. Every settlement we establish becomes a temporary barrier — the creep consumes the organic substrate we've established and then continues. We are not the problem. But we occupy the problem's path."*

Three relocations. Each time, Elena's colony left behind everything they'd built. Each time, they built less, moved fewer people, held less ground. The trajectory was clear: retreat until there's nowhere left to retreat to.

Cole looked at the geological survey data from the original viability report — the same report that had inflated the habitability score by twelve points, the same one with the redacted appendix. The data was compromised, but the topographic mapping was harder to fake. Four kilometers east, the elevation rose. Rock substrate. Mineral-rich ground with less biomass accumulation. If the creep was a decomposer, it would struggle where there was nothing to decompose. The colony could relocate to higher ground, build on stone instead of soil, strip the organic matter from the perimeter.

It was possible. Elena had relocated three times to lower ground, chasing organic soil for agriculture, never breaking the pattern. Cole could break it. Build where nothing grew. Trade agricultural advantage for defensibility.

It was also possible that he was wrong, the way Elena had been wrong three times before she stopped writing.

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Cole brought the plan to Ren.

"We have eleven weeks to the core shelters," he said. "Maybe six if it accelerates the way Elena's data suggests. But Ren — the agricultural zone is in the path. Seventy days, maybe less."

"I know," Ren said. She had her own copy of the photographs, her own projection lines. She'd been running the same numbers. "Once we lose the food production zone, the timeline accelerates. Shortages, morale collapse, decision-making degrades. We move before Day 70 or we move under emergency conditions."

"The creep isn't even at full speed yet. If the second cycle intensifies—"

"It will. And we'll be in the process of relocation when it does, carrying everything we've built on our backs, moving a population that's increasingly malnourished and frightened. Cole, I'm not being pessimistic. I'm reading the pattern." She laid out the timeline on a map, marked the creep's current position and projected advancement. "We have two weeks to decide. Four weeks to plan. Eight weeks to move everything. After Day 70, it's emergency extraction, and we lose infrastructure."

Cole understood what she wasn't saying: we lose the people who can't move fast.

"I scouted the high ground east of here," Cole said. "Rock substrate. Higher elevation. Less organic material for the creep to feed on. Four kilometers. The topographic data in the viability report shows cleared platforms near the drainage features — enough space for shelter construction. Water access from the natural drainage."

"You're talking about starting over."

"I'm talking about surviving."

"Those aren't the same thing," Ren said. "Half this colony will refuse. They've built here. Planted here. They know this ground. You're asking them to walk four kilometers into terrain they've never seen and trust that the math works."

"The math does work."

"Then bring it to a vote."

Cole didn't answer immediately. Ren watched him, and he could feel her reading the hesitation. He didn't trust a vote. Not because he didn't trust the colonists — because he didn't trust fear to produce good decisions. People under survival pressure voted for the familiar. They voted to protect what they had, even when what they had was already dead. A vote on relocation would be a vote on whether to grieve the settlement they'd built, and grief didn't follow timelines.

"A vote takes time we don't have," Cole said.

"Then what are we surviving for?" Ren said. "If you relocate thirty-eight people without their consent, you're not leading a colony. You're running a prison with better scenery."

The words landed. Cole felt them in the same place he felt the math — the place where logic and something harder to name shared space.

"Call the vote," he said.

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Ren organized it in two days. Full colony assembly. Every colonist present, every voice counted.

Cole presented the data. The creep's advance rate. The agricultural zone timeline. The higher ground, the rock substrate, the four-kilometer walk. He showed them the map. He showed them Elena's containment

data — fire's failure, chemical adaptation, the pattern of retreat. He didn't tell them how Elena's story ended. He told them enough.

Half the colony pushed back. Not Polk's faction — Polk surprised everyone by staying quiet, arms crossed, listening. The resistance came from the people who'd built the settlement. The water system engineers, the shelter constructors, the agricultural teams who'd spent eight months learning the soil. They'd made something here. It worked. Walking away felt like admitting the planet had won.

A woman named Kira, who'd designed the water collection system, stood up. "We adapt. We've been adapting since Day 1. We figure out the creep the way we figured out the soil cycle, the way we figured out the rationing. We don't run."

Dana answered before Cole could. "The creep adapts faster than we do. Fire accelerated it. Chemicals bought six weeks. Every strategy we deploy, it routes around. This isn't a problem we solve from here. It's a problem we survive by not being here."

The debate went for three hours. Cole watched it happen and said nothing after his initial presentation. Ren moderated. Voices rose. People argued about shelter construction time, about agricultural restart costs, about the risk of moving malnourished colonists four kilometers through hostile terrain. About Marcus Reeves and what the creep had done to his shelter and whether the perimeter was safe for anyone sleeping within two hundred meters of the forest edge.

The vote was close. Twenty-one to seventeen, with Ren abstaining as moderator.

Twenty-one to relocate. Seventeen to stay and fight.

Cole looked at the seventeen. He knew the math on what staying meant. He also knew that seventeen people had just been told their judgment was overruled by four votes, and they'd be carrying shelter beams across four kilometers of alien terrain for a plan they didn't believe in.

Polk had voted to move. When it was over, he found Cole outside the assembly shelter.

"I'm done watching this place take pieces of us," Polk said.

Cole nodded. Polk walked away. Cole stood in the dark and listened to the creep advancing somewhere beyond the perimeter markers, patient and slow, and thought about Elena's three relocations and the six people she had left when she made the third one, and the fact that Elena had been smarter than him, more experienced than him, and it hadn't been enough.

He had to be enough. Twenty-one people had voted that he was. Seventeen had voted that he wasn't. Either way, they were all moving east in the morning.

# Chapter 12 — Democracy

The vote had been twenty-one to seventeen, and Ren could feel the seventeen in every interaction for the first forty-eight hours after it passed.

She'd pushed for the vote because she didn't trust Cole to relocate thirty-eight people on arithmetic alone. Cole had resisted because he didn't trust fear to produce good decisions. They'd both been right. The vote happened, the margin was four, and now seventeen colonists who'd voted to stay were being asked to dismantle the settlement they'd built with their own hands and carry it four kilometers east because twenty-one people and a fungal network had overruled them.

Ren took the close margin as her signal to get detailed. Speed and structure were the only tools that would keep the seventeen from calcifying into permanent opposition.

She organized the relocation in six days, which meant planning it backward from Cole's timeline and forward from what seemed physically possible. Thirty-eight colonists, limited equipment, four kilometers of terrain between the old settlement and the new site on the high ground. The northern agricultural zone had to be harvested before the creep reached it. The shelters had to be dismantled and transported or rebuilt. Water systems had to be reestablished. Medical supplies had to be moved with redundancy — can't afford contamination or loss mid-transport.

Ren divided the colony into work teams. She didn't ask permission. She presented the structure to Cole, who nodded, and then she presented it to the colony as fact rather than suggestion. The seventeen who'd voted to stay didn't refuse to work — that would have meant refusing to survive — but the resistance lived in the small hesitations, the slower pace, the silence where collaboration should have been. Ren watched it and decided the only solution was making the work so immediate and so physical that grievance didn't have room to breathe.

The problem was that knowledge and execution were different objects.

Day One of the relocation: the agricultural team harvested the northern plots. Thirty-eight people working in the shadow of the creep's advance, pulling crops with the knowledge that this was the last harvest from this soil. The work was physically brutal and emotionally strange — mourning a garden while trying to feed your survival. Ren worked the southern plots with five others, moving methodically through the plants Dana had identified as most likely to survive transplantation. Dana was there too, hands deep in the soil, knowing exactly which species had taken root well enough to be worth moving, which had to be left.

They'd known each other for ten months. In that time, Ren had watched Dana move through the colony with the precision of someone who'd calculated exactly how much space she was allowed to occupy. Quiet. Necessary. Present but not imposing. Ren had developed a respect for that kind of economy — the refusal to waste motion or words. She'd also recognized it as a survival technique, the body language of someone who'd learned early that taking up room was dangerous.

"We're losing three-quarters of this," Dana said, pointing at the transplant pile. "The soil composition at the channels is different. Rock-based substrate, higher mineral content. The plants that thrived here might not establish there."

"Then we take the seeds," Ren said. "Learn to grow what the new soil wants."

Dana looked at her like Ren had suggested something philosophical. "We're not adaptable enough for that. We think we are. We think we're survivors. We're just people carrying what we know to a place that doesn't want it."

Ren didn't argue. Dana was right in a way that stopped argument. They just moved plants, chose which ones to carry, which ones to leave for the creep to process into whatever the planet needed next.

That night, the first death came not from the relocation but from the thing that preceded it.

Thomas Chen, age 51, construction worker who'd been managing the shelter-deconstruction team, had a cardiac event while supervising the removal of the eastern shelter beams. Haruki found him on the ground, respiration shallow, the kind of happening that in a proper hospital would have been reversible with the right intervention delivered in the right time window. Haruki had neither. He had morphine and a shelter and the knowledge that Thomas's heart had decided to stop during the last moment of the relocation's early stage, when supplies were stretched thin and attention was divided.

Cole sat with Thomas. Ren watched from a distance, not intruding but close enough to see Cole's face, which remained neutral in the way that people's faces became neutral when they were doing math inside their heads. By the time Haruki closed Thomas's eyes, Cole had already subtracted him from the thirty-eight and begun recalculating margins.

The count was thirty-seven.

Day Two and Three: the physical relocation began in earnest. Ren coordinated the transport teams, pacing them to avoid exhaustion-based accidents, rotating roles to prevent muscle strain from becoming injury. She moved between the teams, watching for signs of failure. A colonist named Patricia developed a cough during the water system relocation — not alarming yet, but Ren noted it, marked it as a potential future complication.

On the night of Day Three, the temperature dropped eleven degrees in two hours. They'd learned to expect the swings from the second cycle, but not while moving across open terrain with no shelter overhead. A colonist named Noor — forty-three, electrician, one of the seventeen who'd voted to stay — collapsed during the night transport. Hypothermia compounded by dehydration and three days of physical labor on reduced rations. Haruki found her unresponsive behind the carrying line. He worked on her for forty minutes in the dark while the transport team stood around him holding crates they couldn't set down on open ground. She didn't come back.

Thirty-six.

Ren redistributed Noor's transport load, adjusted the schedule to include mandatory thermal checks every hour, and kept the teams moving. There was no time to stop and no shelter to stop in — they were between settlements, exposed, carrying everything they owned across four kilometers of terrain that belonged to something other than them. The seventeen who'd voted to stay said nothing. Noor had died doing the thing she'd voted against, and the math of that was worse than the cold.

Day Four and Five: the transport continued. Ren watched the remaining colonists for signs of breakdown — physical or otherwise — and saw both. People moved slower. Spoke less. The relocation was bleeding them through accumulated stress the way the creep bled the forest through patience.

On Day Five, during a night transport run, one of the carrying teams encountered terrain they'd misjudged. The equipment manifest listed seventeen items as critical — medical supplies, water filtration components, seed stock for crops. James was carrying one of the backup water filtration units, walking a path that had seemed navigable in daylight but had transformed into something treacherous in the darkness. His foot caught. His balance failed. He fell, and the filtration unit fell with him, and his right leg bent in a way that legs shouldn't bend.

Haruki splinted it in the field. The fracture was compound and dirty, which meant infection was not just possible but approaching inevitably. They carried James to the new settlement, and Haruki did what he could — cleaned the wound with boiled water, set the bone as well as field medicine allowed, wrapped it

immobilized. James didn't cry. He just lay in the medical shelter and did the same math Cole was doing, which was the calculation of whether his utility was greater than the resource cost of keeping him alive.

"He'll be functional," Haruki told Cole and Ren, using the medical term that meant mobile enough to contribute but disabled enough that he'd need support. "In three weeks, maybe four. The infection risk is moderate if we manage it carefully."

The phrase "if we manage it carefully" was Haruki's way of indicating that careful management was his priority but not guaranteed. Ren understood. Cole did too.

On Day Six, they completed the initial transport. The old settlement was stripped of everything portable and useful. The new site at the channels was beginning to look less like a refugee camp and more like a settlement — the shelters reassembled with careful attention to position, the agricultural plots staked out in places Dana had identified as having suitable soil composition, the medical shelter established as a secondary priority after water and food systems.

That was when the work team clearing the new site discovered the channels had structure.

Ren was coordinating the final supply transport when one of the excavation team ran over, breathless with discovery rather than panic. "You need to see this," he said. His name was David, mid-forties, geologist on Earth though his job here was general construction. He had the expression of someone who'd encountered something that rearranged his understanding of the problem.

The channels were deeper than they'd realized. The surface cut ran five meters below the soil line, but when David's team started clearing rock for shelter foundations, they found that the channels continued further down. More than that: they weren't natural erosion patterns. The walls were too regular. The substrate was too uniform. And the mineralized substance that filled the deepest sections wasn't random accumulation — it had geometric structure, almost like crystalline growth, or deliberate storage.

Ren crouched at the edge of the excavated section and looked at what remained of an infrastructure that had been built by something that wasn't human and wasn't recent. The channels were carved, not eroded. The mineral deposits were arranged, not scattered. The system was, by any measure that mattered, intentional.

"How old?" she asked David.

"I can't tell. Geologically, it could be anything from five hundred to five hundred thousand years. But it's old enough that the rock has partially reabsorbed the carved sections — there's weathering, mineral deposition, the kind of slow integration that happens over long timescales. Nothing here is recent."

Ren stood and looked at the empty space where the ancient settlement had been. Whatever had lived here once had built infrastructure. Water management systems, at minimum. Possibly agricultural. Possibly something else entirely — a system that had no Earth analogy, designed for needs that Earth minds couldn't imagine.

She walked to the medical shelter where Cole was checking on James's condition. Cole looked up when she entered, reading her face the way he'd learned to read hers — for signs of crisis or discovery.

"The channels aren't natural," she said. "They're architecture. Something built here. A long time ago."

Cole was quiet for a moment. Then: "How long?"

"David doesn't know. Old enough that it's been integrated into the geology. Thousands of years, probably. Maybe longer."

Cole stood and walked to the shelter entrance, looking out at the new settlement and the channels beyond it. The evening was coming on, and the sky above Parcel 16b was shifting toward its greenish tint. The trees hummed, the sound that had become so familiar Ren barely noticed it anymore.

"We're not homesteading," Cole said. It wasn't a question.

"No. We're inheriting."

Cole nodded like that was a calculation that fit into some equation he'd been running. Maybe it did. Maybe it changed everything about the math, or maybe it didn't change anything. Maybe either way, there were thirty-six colonists alive and one whose compound fracture was already incubating the kind of infection Haruki couldn't guarantee he could manage.

The creep hadn't followed them up. Ren had walked the eastern perimeter that morning — the rock substrate was clean. Whatever the fungal network needed, it wasn't here. The higher ground was defensible not because they'd outrun the thing, but because the planet's own decomposition system stopped where there was nothing left to decompose. One problem solved by geography. The rest unsolvable by anything except time.

Ren thought about Thomas Chen's cardiac event on Day One, about Noor cooling on open ground while the transport team stood in the dark holding crates they couldn't set down, about Patricia's cough, about James's leg and the infection already building in the dirty wound, about the thirty-eight who'd become thirty-six in six days, about the system this planet had built for its own renewal and the people they'd buried in its soil trying to establish a foothold.

She thought about democracy and whether it meant anything when the margin was four votes and everyone was already dying.

"We need to assign research teams," Ren said. "Figure out what the channels were built for. If there's infrastructure, there might be systems we can use. Water management, at minimum."

Cole looked at her. "That assumes the builders solved the same problems we're trying to solve."

"They lived here. They built. They survived something. If they didn't, this infrastructure wouldn't exist."

"They're gone," Cole said. "Whatever they were, they're not here. We have thirty-six people and a planet that changes the rules every orbital cycle. Channels are interesting. They're not food, shelter, or medicine."

"They might be all three," Ren said. "We survive by understanding the system we're in. That system includes infrastructure someone else built. We study it. We learn. We use what's useful."

Cole didn't argue further. Ren understood that as agreement — Cole was too practical to ignore a potential resource, even if his instinct was to minimize distraction. She'd push the research forward, and Cole would allocate people to it, and they'd find out what the channels meant. Or they wouldn't. But at least they'd look before the planet decided they weren't worth the space.

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**Author Notes — Chapter 12**

**Voice Decisions:** Ren's POV provides the external perspective that Cole's interiority lacks. She sees the relocation as both necessary and fragile — the work is brutally practical, but it's also preparation for a failure Cole already knows is coming. The chapter's tone is darker than 11 because Ren's observational precision catches the death count without Cole's calculation shield. Thomas Chen, Noor, and James's injuries come without warning because exhaustion and stress don't announce themselves — they just happen. Noor's death — one of the seventeen who voted against relocation — adds political weight to the body count: she died doing the thing she'd voted against.

**Structural Notes:** The relocation spans Days 1–6 as a compressed sequence (summary with scene punctuation rather than full scene detail). The chapter moves from action → death → discovery → implication, ending with Ren and Cole recognizing that the architecture changes the problem without changing its fundamental pressure. The vote isn't re-dramatized because Chapter 11 already delivered it; Chapter 12 is about the cost of execution. The creep's failure to follow onto rock substrate provides relief without resolution.

**Continuity:** Vote was 21-17 (close, per outline), carried forward from Ch 11. Count moves 38 → 37 (Thomas Chen, cardiac event, Day 1) → 36 (Noor, exposure/hypothermia, Night 3). James's compound fracture and infection risk established (Day 5). Patricia's cough introduced as a future complication (will develop in Ch 16). The ancient channels are now identified as deliberately built, not natural. Creep confirmed as not following to rock substrate. The new site is operational by chapter's end.

**Off-Voice Flags:** Cole's dialogue about the channel builders tightened from earlier draft — removed the enumeration ("consumed them or they left or they failed") in favor of a more direct, characteristically terse dismissal ("Channels are interesting. They're not food, shelter, or medicine."). The pragmatic-dreamer tone holds throughout.

**Word count:** ~3,200 words

# Chapter 13 — The Structure

The channels went deeper than anyone had expected.

In the two weeks since the relocation, David's excavation teams had been clearing rock for shelter foundations and uncovering more of the network that ran beneath the settlement. What had looked like drainage cuts on the surface turned out to be the top layer of a system that extended at least twelve meters into the bedrock. Cole stood at the edge of the deepest excavation trench on Month 11, Day 4, looking at walls of carved stone filled with a substance that caught the light like smoky quartz.

Dana had been studying it since the discovery. She'd collected thirty-seven samples from different points in the network, mapped the mineral composition at each depth, cross-referenced the crystalline structure against her creep data. The results came back on a morning when the sky was doing its green-shift thing, the light falling through the spiral trees at an angle that made the channel walls glow faintly from within.

"Same compound," Dana said. She was crouched at the edge of the trench, holding a sample container up to the light. Her voice had the quality it got when data confirmed something she'd suspected but hadn't wanted to be right about. "The base molecular structure is identical to the creep. Same organic-mineral hybrid. Same filament architecture at the microscopic level."

Cole looked at the substance in the container. Inert. Glassy. Nothing like the slow-moving biological mat that had driven them from the original settlement.

"It's not active," he said.

"No. It's ancient. Mineralized. The organic components have been replaced by mineral analogs over time — the same way petrified wood preserves structure after the original material is gone. This is what the creep becomes. Or what it was. The timeline runs in both directions." Dana set the container down carefully. "Cole, the creep isn't a disease. It's a phase. The channels are full of the end product. The planet builds these networks the way coral builds reefs — living processes that create mineral infrastructure over deep time."

Cole processed that. The creep they'd fled from was the active version of the substance lining the walls beneath their feet. The ancient builders hadn't carved channels and then filled them. They'd used the creep — or something like it — to build.

"How old?" he asked.

"I can't tell precisely. David's geological estimates put it at tens of thousands of years minimum. The mineralization is complete. Whatever was alive in these walls hasn't been alive for a very long time."

"But the builders were alive. Something lived here that understood how to use this."

Dana stood. Brushed the rock dust from her knees. "Something lived here that built infrastructure on a planetary scale. The channels aren't local. They branch. Every time David's team clears a new section, they find more connections. This isn't a settlement. It's a network. Drainage, nutrient transport, maybe communication. Maybe something we don't have a word for."

Cole looked at the trench. The carved walls. The mineralized substance that had been deposited with geometric precision into rock that was older than human civilization. He felt the weight of it — not reverence, not awe. Something more practical. The recognition that every assumption the colony had made about this planet was built on a surface reading. Below the surface, the planet had a history, and they were standing on it without knowing the first chapter.

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The colony took the news in pieces.

Ren organized a presentation. Dana delivered the scientific data. David showed the excavation maps, the branching network, the depth measurements. Cole stood in the back and watched thirty-five faces absorb the information that the ground beneath their shelters was architecture.

The reactions split along lines Cole had learned to predict. The practical contingent — the builders, the water engineers, the people whose survival instincts ran toward the tangible — wanted to know if the channels could be used. Water management. Drainage. Storage. If the ancient builders had solved the problems the colony was still fighting, maybe their infrastructure was the answer.

The philosophical contingent — smaller, quieter, harder to identify until they spoke — wanted to know what this meant. Who had been here. What had happened to them. Whether the colony had any right to build on top of something they didn't understand.

Garrett Polk, characteristically, occupied both camps at once. "Can we drink from the channels?" he asked. Then, before Dana could answer: "And what killed the things that dug them?"

Dana answered the first question. "The water flow through the upper channels is clean. Mineral-rich but potable. We've been using it already — the drainage features David identified are part of this system. So yes. We've been drinking from the channels since we arrived."

She didn't answer the second question. Nobody could.

Cole's interiority during the presentation was simpler than anyone watching him would have guessed. He ran the math. The channels existed. The channels carried water. The channels provided defensible infrastructure on rock substrate the creep couldn't colonize. Did the channels change the survival math? No. The colony was at thirty-six, the same thirty-six it had been since Noor died on Night Three of the relocation. The channels didn't add people. They didn't subtract threats. They provided resources the colony was already using without knowing what they were.

Did the channels change the moral math? Cole didn't let himself answer that. The moral math was Ren's territory. She'd fight it out in council, argue about inheritance and obligation, push for research teams to study the builders' legacy. Cole would let her. Not because he agreed or disagreed, but because the moral questions didn't change the number. Thirty-six. Twenty-four needed. Forty-eight months remaining.

Elena's logs never mentioned this. Her colony had built twelve kilometers west, on different terrain. She'd found the same creep, the same orbital cycles, the same slow death. But she'd never stood over a trench cut into bedrock and realized the planet had been inhabited before. That absence in her data was its own kind of information. The planet was large enough to contain different stories. Elena's and Cole's were parallel but not identical. The channels were his alone.

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Ren pushed for research teams. Cole allocated three people: Dana for the biological analysis, David for the geological survey, and a rotating fourth position for general assistance. He kept the teams small because every person studying channels was a person not building shelters, not tending the dormancy-phase crops, not maintaining the systems that kept thirty-six people breathing.

Suri Park volunteered for the fourth position on the first day.

Cole almost said no. Not because Suri was unqualified — she'd been working as Dana's field assistant since the fruit incident in Month One, when she'd eaten the native berries that nearly killed her and then come back the next week asking Dana to teach her how to test samples properly. Suri had the curiosity that made people useful and the eagerness that made them careless, but eight months of working with Dana had filed the carelessness down to something closer to controlled intensity. She asked questions before she touched things now. She waited for data before she acted.

Cole said yes because Suri had earned it. She'd survived the thing that should have taught her to be afraid, and instead it had taught her to be precise. He respected that more than he respected caution.

The research moved quickly. Over six weeks, the team mapped the channel network beneath the settlement and extending two hundred meters in each direction. The branches were systematic — main arteries splitting into secondary channels, secondary channels splitting into capillary networks so fine they could only be detected with Dana's ground-penetrating analysis. The geometry was consistent. The mineralized compound filled every channel to a uniform depth, as though the entire network had been activated simultaneously and then gone inert at the same moment.

Dana's working theory: the channels were a planetary circulatory system. During active orbital phases, living creep — the biological mat — would flow through the network, transporting nutrients, breaking down dead matter, redistributing minerals. During dormant phases, the compound would mineralize, preserving the infrastructure for the next cycle. The builders hadn't fought the planet's cycles. They'd built with them. Used the creep as a construction material, a transport medium, a living tool.

"It's elegant," Dana said one evening, spreading her maps across the research shelter's table. Cole, Ren, and Suri were there. The light outside was fading, the green-shifted sky darkening toward something that wasn't quite night. "The planet has a circulatory system. The builders understood it. They didn't try to stop the cycles or survive them — they integrated. The channels are infrastructure that works with the planet instead of against it."

"Infrastructure that's been dead for tens of thousands of years," Cole said.

"Dormant," Dana corrected. "The compound mineralizes during dormancy. That doesn't mean the system is permanently inactive. It means it's waiting."

Cole heard the word and filed it. Waiting. The channels were waiting. The compound that had killed Marcus Reeves, that had driven them from the original settlement, that had consumed Elena's colony centimeter by centimeter — that substance, in its ancient form, was sitting in the walls beneath their shelters. Dormant. Mineralized. Waiting.

"Waiting for what?" Ren asked.

"For the cycle to deepen enough to reactivate it. The surface creep is the active phase. What's in the channels is the stored phase. If the orbital mechanics produce a deep enough cycle — one that goes past what we've already seen — the stored compound could theoretically reactivate."

"Theoretically," Cole said.

"Based on the data I have. The mineralization in the deepest channels is complete. Those sections haven't been active in geological time. But the shallower channels — the secondary and capillary networks closer to the surface — the mineralization is less complete. Younger deposits. More recent cycles."

Cole looked at the map. The deep channels, fully mineralized, ran like bones beneath the settlement. The shallower ones branched upward, closer to the surface, the compound less glassy, more granular. Transitional. The deeper you went, the older and more inert the material. The shallower you went, the closer it was to the thing that still moved through the forest.

"We need to map the activation thresholds," Suri said. She was leaning over the table, tracing a capillary branch with her finger. "If we can determine what conditions trigger reactivation in the shallow channels, we can predict when sections become dangerous."

It was a good question. Cole recognized it as the kind of question that came from someone who'd learned to think in systems — not just what is happening, but what would cause it to happen differently.

"Agreed," Dana said. "I'll design the testing protocol. We'll need ground temperature data, mineral composition at various depths, and samples from the transitional zones — the sections where full mineralization hasn't occurred."

Cole approved the expanded research. Allocated one additional day per week for deep-channel exploration. The colony could afford it. The dormancy-phase crops were established, the water systems were running through the channels' own drainage, and the creep hadn't followed them to the rock substrate. For the first time in months, Cole felt something that wasn't dread. Not optimism — he didn't trust optimism. Something more structural. The sense that the problem had a shape he could see, and the shape had edges he could work with.

The channels were a system. The creep was a phase of that system. The planet operated on cycles that could be mapped, predicted, integrated. Dana had cracked the soil. She was close to cracking the creep. The builders had already solved this problem, thousands of years ago. Their solution was carved into the bedrock. All the colony had to do was read it.

Cole believed that. Standing over the maps, looking at the elegant branching network, listening to Dana explain the circulatory logic, watching Suri trace the capillary paths with her finger — he believed they were close to understanding. The planet wasn't hostile. It was a system. Systems could be learned.

That belief was the most dangerous thing Cole had carried since landing.

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Month 12, Day 8. Cole, Dana, Suri, and Ren descended into the deepest accessible section of the channel network.

The excavation team had opened a shaft two weeks earlier — a natural fracture in the bedrock that intersected one of the primary channels. The channel was wide enough to walk through at the upper levels, narrowing as it descended. The walls were lined with the mineralized compound, the surface catching their lamplight in slow ripples of smoky translucence. The air was cool. Dry. It smelled like stone and something faintly metallic, the same trace mineral scent that lived in the water they'd been drinking.

They'd explored this section three times before. Each time, the compound had been inert. Glassy. Cold to the touch. Dana had taken samples from the walls at every depth, tested conductivity, moisture content, thermal response. Dead material. Museum-quality preservation of something that had once been alive, now serving only as infrastructure.

Cole led. Ren followed with the mapping equipment. Dana stopped at intervals to sample the wall composition, noting changes in the mineralization gradient. Suri moved between them, documenting, asking questions, pressing her palm flat against the channel walls the way she'd started doing on the second expedition — feeling for temperature variations, moisture, anything the instruments might miss.

"The gradient is shifting," Dana said at the seven-meter depth mark. She was studying a sample under portable magnification, the light from her headlamp pooling on the container. "The mineralization is less complete here. See the granular structure? This section is younger. Deposited during a more recent cycle."

Cole looked. The compound in the walls did look different — less glassy, more textite. The surface had a faint roughness, like sugar that hadn't fully dissolved.

"Recent meaning what?" he asked.

"Hundreds of years instead of thousands. Maybe less. The transition between full mineralization and partial is gradual. We're moving down through time layers — the deepest channels are the oldest, the shallower ones are newer."

"We're going deeper," Cole said. "So we're going older."

"In the main channels, yes. But the capillary branches that intersect the primaries can be any age. Some were deposited recently — within the current set of orbital cycles. The planet keeps building."

They continued down. The channel narrowed at the nine-meter mark. Cole had to turn sideways to pass a rock shelf that jutted into the passage. Beyond it, the channel widened again into a junction — two primary channels meeting at an angle, the walls thicker with the mineralized compound, the geometric precision of the carved stone visible where the coating had cracked and fallen away.

Suri moved ahead of Cole into the junction. She did that — moved toward the interesting thing, the next data point, the question that was forming in her head. Dana had trained the recklessness out of her, but the curiosity remained. It was why she was useful. It was the thing Cole valued.

"The walls are different here," Suri said. She was pressing her palm against the junction wall, the way she always did. "Warmer. Not warm, but warmer than the upper sections. Can you feel it?"

Cole placed his hand on the wall. She was right. A few degrees above the ambient rock temperature. Not alarming. Consistent with geothermal activity at depth. He noted it.

Dana was ten meters back, still sampling at the gradient transition. Ren was between them, recording the junction's dimensions.

"I want to check the capillary branch," Suri said, pointing at a narrow channel that split off from the junction at a downward angle. It was barely wide enough for a person. The mineralized compound on its walls was visibly different — more granular, less glassy, the surface rough under the lamplight. A younger deposit. Closer to active.

"Take a temperature reading first," Cole said.

Suri pulled the thermal probe from her kit. Held it against the capillary wall. "Four degrees above ambient. Consistent with what we've been seeing. No anomalies."

"Go ahead. Stay within voice range."

Suri entered the capillary branch. Cole watched her lamplight shrink as she moved down the narrow passage, her shoulders brushing the walls, the light catching the granular compound on either side of her. She was taking samples as she went — scraping the wall surface into collection vials, labeling them, methodical.

Cole turned back to the junction. He was examining the geometric patterns in the carved stone — the precision of the angles, the uniformity of the channel width — when the air changed.

Not temperature. Not smell. Something more fundamental. A vibration, subsonic, the same frequency range as the humming trees but concentrated, directional. It came through the rock. Through his hands where they rested on the junction wall. Through the soles of his boots. The channel was resonating.

Dana felt it too. She was on her feet, moving toward him, her face carrying the expression she wore when data reorganized itself faster than she could process.

"Cole," she said. "The thermal readings just spiked. The capillary networks. All of them. Simultaneously."

He turned toward the capillary branch where Suri had gone. The lamplight was still visible, thirty meters down the narrow passage. Suri's voice came back: "The walls are changing. The surface — it's getting soft. Like the mineralization is reversing."

Cole was already moving. The passage was tight. His shoulders scraped the walls. The compound on either side of him — the glassy, inert, ancient material — was warm now. Not geothermal warm. Active warm. The

surface had developed a sheen, a faint moisture that hadn't been there ten minutes ago. The mineralization was reversing. The stored phase was waking up.

"Suri, come back. Now."

"I'm coming. The floor is — Cole, the floor is moving. There's flow. It's thin but it's moving."

He could see her lamplight bouncing as she turned. Twenty meters between them. The capillary channel was too narrow for him to pass her. She had to come to him.

Then the vibration deepened. The walls pulsed — not metaphorically, not in his imagination. The compound expanded. The channel narrowed by centimeters. The floor, which had been solid rock coated with mineralized compound, was now coated with something that had the consistency of wet calcium. Flowing. Slowly. The dormant channels reactivating in response to the deepening cycle, the same way the surface creep had activated months ago, but down here the compound was concentrated, old, and dense.

Suri was moving toward him. Fifteen meters. She was fast. She was precise. She was doing everything right.

The compound on the floor was rising. Not like water — like something growing. The filaments emerging from the mineralized base, the same lace-like structures Dana had documented in the surface creep, but denser. Older. Fed by whatever reserves the ancient network had stored during its long dormancy.

"Cole." Suri's voice wasn't a scream. It was confusion. The tone of someone encountering a variable that didn't match the model. "It's on my boots. I can't — it's dissolving the soles."

He could see her now. Twelve meters. She was walking, not running, because running in a narrow channel with dissolving footing was how you fell, and falling was how you died. She was being precise. She was thinking.

The compound was at her ankles.

Cole pushed deeper into the capillary. His own boots met the flow — thin, warm, the surface of the rock slick with reactivated compound. He felt the dissolution start on the rubber soles. The filaments were indiscriminate. Marcus Reeves. The same process. Except Reeves had been asleep and this was happening in real time, in a channel barely wide enough for one person, with Suri ten meters ahead of him and the gap between them full of something that was waking up faster than either of them could move.

"Suri. Keep coming."

"I am." Her voice was controlled. Tight. She was looking down at her feet, at the compound climbing her boot shafts. She took another step. Another. Eight meters.

Cole reached for her. The channel was too narrow. He couldn't get past the rock shelf that jutted into the passage. His arm extended into the space, his hand open, and Suri's hand reaching back, and between them the compound flowing, rising, the ancient system cycling into its active phase in a section that had been dormant every time they'd walked through it.

Suri's feet stopped moving. The compound had reached her shins. She looked down. Looked at Cole. Her face held the expression of someone doing a calculation that didn't produce a survivable answer.

"I can't feel my feet," she said.

"Suri."

"I know." Two syllables that held everything — the understanding, the mathematics, the impossibility of her position. She reached her hand forward. Cole stretched until his shoulder ground against the rock shelf. Their fingers touched. The compound was at her knees.

The dissolution was fast. Faster than the surface creep. Concentrated. The ancient compound, reactivated after centuries or millennia of dormancy, had reserves the surface version didn't. It moved through Suri's lower body with the efficiency of a system resuming function after a long pause — not aggressive, not hostile. Just operational. Doing what it was designed to do: break down organic material, cycle nutrients, build infrastructure. Suri was organic material. The compound didn't distinguish.

Ren was behind Cole, pulling at his shoulder. "Cole. Come back."

"No."

"You can't reach her. Cole. You can't."

Suri's hand went slack. Her eyes were open but the focus was gone — shock, neural disruption, the same unconsciousness that had been Haruki's single mercy for Marcus Reeves. She was unconscious before the significant dissolution. That was the only comfort available, and it wasn't enough.

Cole let Ren pull him back. They retreated up the channel, boots dissolving, the compound following them at the pace of something patient, and reached the junction as the capillary branch behind them filled with active compound. The main channels — the deep, fully mineralized arteries — remained inert. The activation was localized. A capillary response. A shallow branch waking up because the cycle had deepened past the threshold the younger deposits needed.

Dana was at the shaft entrance, pulling them up. Cole's hands were bleeding where the rock shelf had scraped them. He didn't feel it.

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Haruki went down with a sealed suit and a thermal probe six hours later, after the activation subsided. The compound had re-mineralized. The capillary branch was inert again, the walls glassy, the floor solid. As though nothing had happened.

There was nothing left to recover.

Haruki's report was brief. Same dissolution pattern as Marcus Reeves. Accelerated by concentration and confinement. The compound had consumed everything organic and deposited the minerals into the channel walls. Suri Park was part of the infrastructure now. Built into the system she'd been studying.

Thirty-five.

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Cole sat in his shelter that night with the thermal data from the capillary branch. The readings were clear. The shallow channels — the younger deposits, the sections where mineralization was incomplete — had reactivated when the second orbital cycle deepened past a threshold Dana's models hadn't predicted. The deep channels, fully mineralized, remained inert. The activation was a gradient: the younger the deposit, the lower the threshold needed to reactivate it.

They'd explored that capillary three times before. Each time, the compound had been dormant. Each time, the thermal readings had been normal. The section was safe because the data said it was safe, and the data was a snapshot of a system in motion, and Cole had read the snapshot as the whole picture because he'd wanted to. Because the channels felt like an answer. Because the planet's infrastructure felt like something he could learn, map, predict, use.

He hadn't miscalculated. The channels had been dormant. His readings were accurate. His protocols were sound. He'd sent Suri into a passage that every data point said was safe, and the data points were correct at the time of collection and obsolete by the time she reached the thirty-meter mark.

The planet wasn't hostile. It was incomplete. His understanding was incomplete. Those were the same thing.

He'd been confident. Standing over Dana's maps, tracing the elegant branching network, listening to the circulatory logic, feeling the shape of a problem he could finally see — he'd believed he was close. The builders had solved this. The solution was in the rock. All the colony had to do was read it.

Suri had died while Cole was reading. She'd died in the space between his understanding and the planet's actual complexity. She'd died because Cole believed he'd figured it out, and believing you've figured it out is the moment the system shows you what you missed.

He thought about Elena. Three relocations. Each time, a retreat from something she thought she understood. Elena was smarter than him. More experienced. She'd watched the creep consume her colony and she'd never found the channels, never had the infrastructure data, never had Dana's maps or David's excavations. She'd failed with less information. Cole was failing with more.

The difference wasn't the data. It was the confidence the data produced. Elena had been cautious and died slowly. Cole had been confident and Suri had died quickly. Neither approach protected you. The planet didn't care about your strategy. It cared about its cycles, its systems, its ancient circulatory logic that had been running since before humans existed and would continue running after they were gone.

Thirty-five. The count that matched his worst-case projection minus one. He'd built a model in his head on landing day — a curve of expected attrition, flattening over time as the colony learned and adapted. The model was holding. The model was accurate. Suri Park was a data point on a curve that Cole had drawn before he knew her name, and the fact that the curve predicted her death didn't make it acceptable. It made it his.

Not because he'd been reckless. Because he hadn't been reckless enough. He'd been careful, methodical, precise. He'd read the data. He'd followed the protocols. He'd believed the system was knowable. The planet had shown him otherwise, and Suri had been standing in the space where the lesson landed.

Cole turned off the lamp. Sat in the dark. The hum of the trees pressed through the shelter walls, the same subsonic frequency that ran through the channels, through the rock, through the ancient compound that was dormant now but wouldn't always be.

He didn't count. He didn't calculate. He just sat with the weight of a number that used to be a person, and let it be heavy.

# Chapter 14 — "Signal"

The soil analyzer weighed eleven pounds and looked exactly like what it was supposed to be — a portable spectrometer modified for field use, cased in reinforced polymer with Asterlyne's logo stamped on the base plate. Dana had requested it as part of her botanical equipment manifest before departure. Legitimate. Documented. One of thirty-seven items she'd signed for on the transport vessel, each one cataloged against her research assignment.

The comm array nested inside the spectrometer's housing weighed four ounces. It drew power from the same battery pack. Used the same antenna port. Transmitted on a frequency buried in the spectrometer's standard diagnostic band, indistinguishable from equipment telemetry unless you knew the encoding protocol. Asterlyne's technical team had built it. They'd given it to her three days before launch, in a windowless office on the fourth floor of the onboarding center, the same building where Cole had been reading the two-hundred-page charter one floor below.

Dana powered it on.

The screen was small. Two inches. Monochrome. The text came through in blocks — compressed, stripped of formatting, the kind of communication designed to leave no fingerprint. She'd been receiving and transmitting through this screen for thirteen months. The rhythm was fixed: transmit window every eleven days, aligned with the planet's magnetic cycle, when atmospheric interference dropped low enough for the signal to reach the relay satellite Asterlyne had parked in high orbit. One relay satellite. One frequency. One person on the ground with access.

Her handler didn't have a name. A designation: FIELD-SUPPORT-7. The messages arrived formatted as equipment diagnostics. The questions embedded in the telemetry looked like calibration requests to anyone who didn't know the pattern.

She knew the pattern.

Dana opened the current batch.

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Thirteen months of reports. She could recite the categories without looking: soil composition data by quadrant, crop yield measurements, water quality metrics, caloric intake estimates, weather pattern observations, equipment status and degradation rates. The scientific data Asterlyne had contracted her to collect. Legitimate research outputs that any agricultural specialist would generate. If someone found the transmissions and decoded them, they'd see what looked like a dedicated scientist sending her findings to the company that funded the expedition.

The other categories she didn't let herself recite: colony population status, leadership structure, interpersonal conflicts, resource allocation decisions, morale assessments, individual behavioral profiles.

She typed up the report the way she always typed it up. Data first. Clean. Quantitative. Growing conditions in the relocated settlement during Months 11 and 12 — the rock substrate at the channel site had different thermal properties than the original soil. Crop transplants from the first settlement showed forty percent viability. Native species from Dana's cycle mapping were performing better on the new terrain. Water supply through the channel drainage was reliable. Mineral-rich but potable.

Then the qualitative section. Council governance stable. Ren Tsai's democratic framework operational. Garrett Polk integrated into the council, no longer pursuing rescue signal. Colony morale impacted by Suri Park's death in the channels. Cole Maren maintaining operational command through logistics control.

Dana stopped typing. Looked at the last sentence. Deleted "through logistics control." Retyped: "Cole Maren continuing to manage colony operations."

The difference was nothing. The information was the same. But the first version sounded like an assessment of a vulnerability, and the second sounded like a status report. Dana had become expert at this distinction — the tiny calibrations that let her believe she was reporting data, not enabling action. The line between observation and intelligence was a line she redrew every eleven days, and every eleven days it moved a little further from where she'd originally placed it.

She finished the report. Agricultural data, population count, social dynamics, resource status. Hit the compression protocol. The transmission would go out during tonight's window.

Sixty thousand credits. Medical debt. Her mother's oncology treatment on Earth — the three-year protocol that saved her life and generated a bill that would take Dana fourteen years to pay on a botanist's salary. Asterlyne's offer had been simple: two years of field reports, debt erased. Not reduced. Erased. The kind of number that makes you stop calculating and start rationalizing.

Dana had done both. She'd calculated the odds — forty-seven colonists, five-year charter, the survival projections she'd read in the data appendix that most people skipped. The math said some of them would die. The math said the data she was sending wouldn't change that. Growing conditions and soil composition didn't kill people. Resource allocation reports didn't subtract from the headcount. She was sending information that Asterlyne could get from their own survey data if they bothered to look. She was just making their jobs easier.

That's what she told herself on Day 1. On Day 30. On Day 90. On Day 180, when she transmitted the colony's first death toll alongside her crop yield data and didn't pause between the two entries.

Thirteen months in, she still told herself the same thing. The words hadn't changed. The weight of them had.

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The new questions arrived three days before the next transmission window.

Dana opened the diagnostic batch during her morning soil sampling routine, the spectrometer propped on a rock at the edge of the relocated settlement's southern agricultural plot. The green-shifted sunlight made the screen hard to read. She angled it with her body, blocking the light with her shadow. A scientist checking her equipment. Nothing to notice.

FIELD-SUPPORT-7's questions had always followed a pattern. Agricultural: soil composition, growth rates, cycle projections. Logistical: supply inventory, equipment degradation, energy reserves. Social: leadership dynamics, faction status, morale indicators. Dana had answered all of them for thirteen months. The pattern was familiar. Comfortable, almost.

This month's batch broke the pattern.

*Confirm current settlement coordinates. Provide GPS-equivalent positioning relative to original landing zone.*

*Detail water source locations and flow rates. Include redundancy assessment — single source vs. multiple.*

*Assess structural vulnerability of primary shelters to sustained wind event (40+ kph, 6+ hours). Include load-bearing analysis if available.*

*Identify supply chain dependencies on individual colonists. Which operational functions are performed by single individuals with no trained backup?*

Dana read the questions twice. Then a third time. The spectrometer hummed in her hands, running its actual soil analysis in parallel with the comm protocol, and for a moment the two functions — the real work and the

hidden work — occupied the same space in her hands with a clarity that felt physical.

Settlement coordinates. They'd relocated. Asterlyne's original survey data placed the colony at Parcel 16b's designated landing zone, four kilometers west. The company didn't know where they were now. The channel site wasn't on any Asterlyne map. The question wasn't scientific curiosity. It was positioning.

Water source locations. Redundancy assessment. Single source versus multiple. Not a hydrological question. A vulnerability question. If the colony depends on one water source, how do you compromise it?

Structural vulnerability to sustained wind events. Load-bearing analysis. Not an engineering question. A stress-test question. How much force does it take to break what they've built?

Supply chain dependencies on individual colonists. Single points of failure.

Dana closed the screen. Put the spectrometer down. Picked up a soil sample container and held it with both hands until the tremor in her fingers passed.

She knew what the questions were. She knew what they meant. The shift from data collection to targeting was as clean as a phase change in one of her mineral samples — same substance, different state, different function. Asterlyne wasn't studying the colony anymore. They were mapping it the way you mapped terrain before an operation. Entry points. Pressure points. Break points.

Dana stood in the morning light at the edge of a field she'd planted, holding a soil container she'd designed, looking at equipment she'd modified to betray every person who depended on her work, and she felt the ground shift beneath her the way it always shifted on this planet. Not physically. Categorically.

She could stop. She could stop right now. Destroy the comm unit. Tell Cole. Tell Ren. Accept whatever came next — exile, judgment, the look on Haruki's face when he learned that the person he'd toasted at dinner was feeding coordinates to the company that had left them to die.

Dana put the soil container down. Picked up the spectrometer. Walked back to her shelter.

She would transmit. Not because the rationalization still held — it didn't, and she'd stopped pretending it did somewhere between the question about water sources and the question about structural vulnerabilities. She would transmit because stopping now raised a different set of calculations. If she went silent, Asterlyne would know the asset was compromised. They'd adjust. Send the next expedition with better surveillance, or worse, assume the colony had turned hostile and accelerate whatever timeline the targeting questions were feeding. Silence wasn't neutral. Silence was a signal too.

So she'd give them coordinates. Not the real ones. Close enough to satisfy the format, offset enough to be useless for anything precise. She'd give them water data. Accurate hydrology, vague positioning. Structural vulnerabilities — she'd describe the shelters as stronger than they were. Load-bearing capacity inflated. Wind tolerance overstated.

And the single points of failure question. She'd answer it honestly: no single colonist represented an irreplaceable operational function. The colony had cross-trained. Redundancy was built in.

That was a lie. Dana herself was the single point of failure for agricultural science, creep research, and soil cycle management. She was the answer to the question she was being asked.

She wouldn't write that.

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The creep samples were waiting in the research shelter.

Dana worked through the afternoon the way she worked through every afternoon — precisely, completely, inside the data the way other people lived inside their own heads. The creep compound from the channel

walls sat in sealed containers on her analysis table. Twelve samples, collected at varying depths from the capillary networks that had killed Suri Park six weeks ago.

The work was the only clean thing she had. The rest of her life on this planet was a construction — the careful scientist, the quiet contributor, the person who took up exactly the right amount of space. But the research was real. The data didn't know who was reading it. The samples didn't know what she'd done that morning.

She was close to something. The mineralized compound from the ancient channels and the active creep from the surface shared the same base molecular structure, but the mineralized version had undergone a specific chemical transformation. The organic filaments had been replaced by mineral analogs — petrification, essentially. The process was driven by a catalyst she'd isolated three weeks ago: a mineral compound present in the rock substrate, concentrated in the deeper geological layers.

If the mineralization process could be triggered artificially — if Dana could synthesize the catalyst and introduce it to active creep — the living compound would mineralize. Become inert. Stop advancing. The planet's own biology, turned against itself. Or rather, accelerated. The creep was always going to mineralize eventually. Dana was looking for a way to make eventually happen now.

The math was promising. Her synthesized catalyst showed eighty-percent efficacy in contained samples. Active creep exposed to the compound began mineralizing within hours. The filaments stiffened. The dissolution stopped. The substance transformed from something that consumed organic material on contact to something that sat in a container looking like smoky quartz.

Eighty percent wasn't enough. Field conditions were different from lab conditions. Concentration, temperature, exposure time — all variables she needed to control before she could propose deployment. But the principle was sound. The planet had already solved this problem. The ancient builders had lived with the creep by integrating its cycle. Dana was trying to do something simpler: interrupt it.

She worked until the light shifted. Logged her findings in the research notes she shared with the colony. Separate from the research notes she shared with Asterlyne.

The overlap between the two sets of notes was ninety percent. The missing ten percent was the catalyst synthesis protocol. Dana hadn't transmitted that yet. She'd transmitted the discovery — the mineralization chemistry, the base compound analysis, the theoretical framework. But the specific protocol, the ratios, the synthesis steps — she'd held those back.

Not out of principle. Out of something colder. If Asterlyne had the synthesis protocol, they could neutralize the creep without the colonists. They could send a second expedition equipped with the solution Dana had built from thirteen months of fieldwork, land them on a planet they'd already mapped through her reports, and operate without the messy variable of forty-seven — thirty-five — people who'd signed a charter that included clause 9.4.2.

Dana hadn't framed it that way in her mind until now. She'd told herself she was holding back the protocol because it wasn't ready. Because eighty percent wasn't publishable. Because a responsible scientist doesn't release incomplete data.

She was holding it back because it was the only leverage she had left.

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Cole had started leaving Elena Vasik's log entries in the common shelter.

Not all of them. Selected passages. Printed on the colony's limited paper supply, the ink fading at the edges where the printer's cartridge was running low. He'd begun after the assembly in Month Seven, when he'd told the colony about the previous expedition — fourteen people, same planet, same data, same cycles. He'd given them the summary then. Now he was giving them pieces of the source.

Dana had read every entry he'd posted. Most of the colony had. Elena's voice carried a specific authority — the authority of someone who'd stood exactly where they were standing and hadn't survived it. Her observations about soil cycles confirmed Dana's data. Her weather predictions matched the patterns they'd observed. Her notes on social fracture read like meeting minutes from the colony council.

Tonight, a new entry was pinned to the common shelter's board. Elena's Month 14. Dana read it standing in the doorway, the shelter empty, the rest of the colony at the evening meal she'd excused herself from.

Elena's handwriting — transcribed by whoever had coded the data core — was precise in the early entries and increasingly compressed in the later ones, as though the physical act of recording was consuming resources she couldn't spare. Month 14 was still in the precise phase. Still the scientist. Still measuring.

*Month 14, Day 6. Supply manifest for the second-quarter delivery arrived via comm relay. Contents significantly reduced from the contracted quantities. Protein stores at sixty percent of agreement. Medical supplies at forty percent. Tool replacements absent entirely.*

*I compared the manifest against the contractual minimum. The reductions correspond to categories I've reported in my field updates — categories where I described the colony as self-sufficient or making progress. Protein reduced because I reported successful native food sourcing. Medical supplies reduced because I reported low injury rates during the stable period. Tool replacements absent because I reported equipment in serviceable condition.*

*Asterlyne is adjusting supply manifests based on our reports. The information I send them is being used to calculate the minimum viable support. Not the contracted support. The minimum. Every competence I demonstrate becomes a justification for reducing what they provide.*

*I am making us expendable by proving we are capable.*

Dana read it twice.

The second reading was slower. She went through it the way she went through mineral analysis — breaking each sentence into components, testing the logic, looking for the seam where the observation became the implication.

Elena had been filing field reports. Standard expedition protocol. She'd reported honestly — progress, capability, adaptation. And Asterlyne had used that data to calculate how much less they could provide. Every piece of good news from the surface became a line item subtracted from the supply manifest. The colony's competence was weaponized against them.

Dana's hands were still. She was very good at keeping her hands still.

Elena's entry didn't mention an informant. Didn't use the word spy or asset or betrayal. But the mechanism was there. Elena had filed reports. Asterlyne had adjusted manifests. The reports were the channel through which the colony's own survival data flowed upward and became the justification for withdrawal.

Elena had been doing it through the official comm relay. The standard reporting structure. She hadn't needed a hidden comm unit or a modified spectrometer. The betrayal was built into the protocol itself. Filing a report was filing a report. Doing your job was doing your job. The company didn't need to recruit an informant when the system itself performed the function.

But Asterlyne had recruited one anyway. They'd recruited Dana. Which meant the official reports weren't enough. They wanted a second channel. An asset with specific instructions, specific questions, specific targeting data that the standard reports didn't capture. Leadership profiles. Social fractures. Structural vulnerabilities.

The pattern was older than her. Elena's colony had been consumed by the same mechanism seven years earlier. Different implementation, same architecture. The company built the betrayal into the charter the way they built the insufficient medical kits — not as an accident, not as an oversight, but as a feature. The information flowed up. The support flowed down. The gap between the two was where colonies died.

Dana wondered if Elena had figured it out. If she'd stood in her own common area and read a supply manifest and felt the same recalibration Dana was feeling now — the recognition that the system wasn't broken. The system was working exactly as designed. Elena's "We're going to make it" was the same kind of certainty Dana had carried into her contract with Asterlyne: the belief that the arrangement was adversarial but survivable. That you could work within the structure and come out the other side with the debt erased and the guilt manageable.

Elena hadn't come out the other side.

Dana unpinned the entry. Read it a third time. Pinned it back. Left it exactly as it was. Walked to her shelter in the dark, past the common meal she could hear but didn't join, past Haruki's medical shelter where the light was still on, past the research tent where her creep samples sat in their containers waiting for tomorrow's analysis.

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The transmission went out at 03:40, during the magnetic window. Dana lay on her cot with the spectrometer balanced on her stomach, the two-inch screen casting pale light on the shelter's ceiling. The data compressed. The signal found the relay satellite. The relay satellite found whatever Asterlyne endpoint was listening.

Settlement coordinates: offset by 1.3 kilometers to the northeast. Close enough to look accurate. Far enough to matter.

Water sources: accurate hydrology, imprecise positioning. The channel drainage system described in general terms, nothing that would let someone find the specific inlets.

Structural vulnerability: overstated capacity. The shelters could handle forty-kph winds for six hours. She reported they could handle fifty for eight.

Single points of failure: none identified. The colony had cross-trained. Redundancy was built in.

Four lies nested in thirty pages of accurate data. The ratio felt like something Dana's mother would have appreciated — the oncologist's daughter, administering a precise dose of deception to keep the host organism alive.

Dana powered down the spectrometer. Set it on the floor beside her cot. The shelter was dark. The hum of the trees pressed through the walls, the same frequency she'd been hearing for thirteen months, the one that lived in the chest more than the ears. Outside, the colony slept. Thirty-five people who depended on her soil maps, her crop schedules, her creep research, her analysis of the planet's cycles. Thirty-five people she was saving with one hand and selling with the other.

She was the most useful person in the colony. She was the most dangerous person in the colony. The two facts occupied the same body, breathed the same air, worked with the same soil-stained hands. There was no version of Dana Wen where one existed without the other. The research was real. The betrayal was real. The catalyst sitting in her lab could save every person here, and the data sitting in Asterlyne's servers could make saving them unnecessary.

Sixty thousand credits. Her mother's life. Thirty-five strangers on a planet that recycled everything, including loyalty.

Dana lay in the dark and listened to the hum and didn't sleep.